CASE STUDY

Thirty Days of Exposure Checks: Version and Configuration Evidence from the Lab

A TeamStarWolf case study. Published 4 November 2026; every source checked on that date except two opened on 3 November (sources 17 and 18).

Executive summary

If you own a patch queue, the question every known-exploited listing puts to you is simple: are we running it, and is it the bad build? This study answers that question for a month of real listings, on real software, with the terminal output to show for it. Between 4 October and 3 November 2026 CISA added 34 vulnerabilities to its Known Exploited Vulnerabilities (KEV) catalog [4]. A lab run on 3 and 4 November 2026 built a check for each of the 24 entries that had a container image or could be checked on a local host, plus a 25th plan for Ghostcat (CVE-2020-1938), which the owner asked for and which is not in the window, and, wherever an image existed, ran it against a matched pair of targets on loopback: one build below the vendor's fix and one at or above it [4] [43].

This is the companion that covers the method and the checks that do not have a study of their own. It walks through six exposure checks in depth (WordPress core, the Linux kernel flaw called Fragnesia, SiteGuard Server Edition, the AhsayCBS pair, a Windows Shell Mark-of-the-Web bypass, and ONLYOFFICE Document Server), the six plans that were blocked, inconclusive or skipped, and what the whole run teaches about version evidence. The short version: a version string is where an exposure check starts, not where it ends. Several fixed and vulnerable builds print the same banner. Several flaws are reachable only when one configuration line is set. Public records disagree: NVD and the GitHub advisory give different affected ranges for Strapi, and the vendor, NVD and CISA weigh the WordPress flaw differently [30] [31] [1] [3] [4]. And a workaround changes the configuration while leaving the version a scanner reads untouched (**Demonstrated**) [43].

> Evidence grade: the vulnerabilities, fixed versions and KEV listings in Parts 1 to 3 are **Confirmed** from the vendor, NVD and CISA records; the fixed builds named in Part 4 are the lab plans' thresholds, sourced in each home study; the in-the-wild exploitation write-ups for AhsayCBS are **Reported**; every lab observation is **Demonstrated** with its capture file named; the playbook lessons drawn from them are **Inferred**.

Key facts at a glance

Field	Value
Window	KEV additions dated 2026-10-04 to 2026-11-03: 34 entries in catalog version 2026.11.03 (1,767 entries in all) [4]
Lab coverage	24 of the 34 entries had a container image or a local check (plus a Ghostcat plan outside the window); appliance entries (Ivanti Connect Secure, Cisco ASA/FTD and Secure Email Gateway, Fortinet, PAN-OS, BeyondTrust, Arista VeloCloud, VMware ESX) had no lab target [4] [43]
Covered in depth here	CVE-2026-119977 (WordPress), CVE-2026-46300 (Linux kernel, Fragnesia), CVE-2026-106931 (SiteGuard), CVE-2026-105133 and CVE-2026-105134 (AhsayCBS), CVE-2026-96951 (Windows Shell), CVE-2021-3199 (ONLYOFFICE)
Blocked, inconclusive or skipped	Blocked: CVE-2015-5477 (BIND), CVE-2023-22894 (Strapi), CVE-2026-88779 (NetScaler). Inconclusive: CVE-2016-3081 (Struts), CVE-2026-21589 (Confluence). Skipped: CVE-2026-110936 (Siemens Software Center) [43]
Summarized with a home study	PHP CVE-2026-109618 and CVE-2026-109619, Node.js CVE-2026-111497, ProFTPD CVE-2015-3306, Nextcloud CVE-2026-110447, Ghostcat CVE-2020-1938, Plex CVE-2026-96652 and GitLab CVE-2026-97229 [43]
Exploit code run	No public exploit that changes a system was run. Most checks read a version, a package record, a configuration line or a log; two sent benign behavior probes: a mod_copy copy of /etc/hostname to a marker file on the ProFTPD pair and a public read-only detection script on the Confluence pair [43]
Oldest flaw newly listed	Five CVEs from 2015 to 2023 added on 2026-10-08, among them ONLYOFFICE CVE-2021-3199, more than five years after NVD published it [4] [23]
Latest listing	WordPress CVE-2026-119977: added 2026-11-03, due 2026-11-06, the only due date in this study still open on 4 November [4]
Most common trap	A banner or bundled-library string that reads the same on fixed and vulnerable builds: ProFTPD, Node.js Brotli and GitLab's Ruby (**Demonstrated**) [43]
Cleanup	Every plan's containers and images removed; 14 dangling anonymous volumes and some empty directories were left on the lab host, not re-checked on 4 November [44]

Key findings

- **A version match is not exposure when a precondition applies.** ONLYOFFICE CVE-2021-3199 needs JWT turned on, Struts CVE-2016-3081 needs Dynamic Method Invocation, NetScaler CVE-2026-88779 is exploitable only with SAML configured (the lab plan's reading), and Ghostcat needs an active AJP connector. For the first three the lab saw a version in the affected range (for NetScaler, from the image tag only), and for Ghostcat the connector line itself differed between builds. The precondition settled ONLYOFFICE and Ghostcat; for Struts and NetScaler it was not observed, which is why those plans stayed inconclusive or blocked (**Demonstrated**; the reading is **Inferred**) [43] [23] [28].
- **Banners lie in both directions.** ProFTPD's vulnerable and patched builds both print `1.3.6rc1 (git)`; Node.js 24.22.0 and 24.22.1 both report Brotli `1.2.0`; GitLab 19.4.1 and 19.4.2 bundle the same Ruby string. Only the package manifest, the product version or the product's behavior told them apart (**Demonstrated**) [43].
- **The public record can disagree with itself.** NVD's Strapi CVE-2023-22894 text still reads "through 4.5.5", while the GitHub advisory runs to 4.8.0 [31] [30]. WordPress rates CVE-2026-119977 low on its own advisory, NVD carries an 8.1 High from Wordfence, and CISA lists it as exploited with a three-day clock [1] [3] [4]. A check built on one record alone gets a different answer (**Inferred**).
- **Workarounds do not move the version.** After the PHP2Shell interim limits were applied, the PHP container still sent `X-Powered-By: PHP/8.4.26`, and the WordPress deny rule changed an HTTP status, not `$wp_version` (**Demonstrated**) [43]. A scanner keyed on version keeps flagging a mitigated host, so the workaround needs its own record (**Inferred**).
- **Parsing bugs are exposure bugs.** A Plex version grep also matched the XML prolog and called a fixed server "NOT patched"; an ONLYOFFICE regex matched nothing on minified JavaScript; `pgrep -f -i siteguard` matched the shell running the check (**Demonstrated**) [43].
- **Lab blockers are facts worth publishing.** A public vulnerable BIND image could not be pulled at all because Docker 28 removed the schema 1 manifest format; a Strapi image needed package downloads on first boot; NetScaler CPX needed a licence acceptance and Confluence a licence key to finish setup, both the owner's to give, not a lab script's (**Demonstrated**; **Confirmed** from Docker and NetScaler documentation) [43] [27] [33].

Top recommendations

1. Treat every KEV entry as two checks, not one: the version against the vendor's fixed build, then the configuration line that makes the flaw reachable. Write both down per product before you scan [43] (**Inferred**).
2. Read versions from the package manager or the product's own manifest (`dpkg-query`, `rpm -q`, `version.txt`, `version-manifest.txt`, `wp-includes/version.php`), and use banners only to find hosts, never to clear them [43].
3. Compare versions with a version-aware tool (`sort -V`, `dpkg --compare-versions`, PowerShell `[version]`), and compare against the threshold for your distribution, not another distribution's line [43].
4. Record workarounds separately from patches, with the configuration evidence that proves them, so the version-keyed finding can be closed or suppressed on purpose (**Inferred**).

5. Where a vendor says patching may erase attack traces, collect them first: SiteGuard tells customers to download diagnostic information before updating [11]. CISA marks forensic triage Yes for six of the seven entries whose field this study records (AhsayCBS CVE-2026-105133 is No) [4].
6. When a lab or a scanner cannot run a check, say so in the record and say why. A "blocked" with a reason is more useful than a silent gap [43] (**Inferred**).

Scope and method

This study covers the KEV additions dated 4 October to 3 November 2026 and the lab run of 3 and 4 November 2026. It concentrates on the plans that ran a check and have no study of their own, and summarizes in one paragraph each the checks whose home is another study. Every source in the Sources table was opened on 4 November 2026, except the Horizon3 and Huntress analyses of AhsayCBS, which were opened on 3 November 2026 and not re-opened; their claims stay **Reported**.

Grade	Meaning
Confirmed	A primary source was opened: the vendor advisory or release note, NVD, the CISA KEV feed, a distribution tracker, or the live MITRE page
Reported	A secondary source's account, not independently corroborated (research write-ups, a threat-hunting report)
Demonstrated	Shown in the lab run; the capture file is named
Inferred	This study's reasoning from the record and the lab, labelled where it appears

How the lab worked (**Demonstrated**, plan.json and the capture files for each CVE) [43]: each plan listed setup, test, exposure-check, mitigation and cleanup steps. Containers ran from public images, bound to `127.0.0.1` or placed on an internal Docker network, in pairs that differed by the vendor's fix. Host-level flaws (the Linux kernel, SiteGuard, Windows) were checked read-only on the host itself. A plan whose only public proof of concept could change a system (create an account, plant a web shell, rewrite a binary in memory) recorded that code and ran a version check instead. The screenshots are renderings of the captured terminal output with a neutral `lab$` prompt and the lab host anonymized. Exploit-class steps are described in one sentence and never shown or quoted.

Part 1: the thirty-day list

What CISA added

The KEV feed (catalog version 2026.11.03) lists 34 additions in the window [4]. The ones this study touches, in order of listing:

KEV added	Due	CVE	Product	CWE (KEV)	Ransomware use	Forensic triage	Lab outcome
2026-10-04	2026-10-07	CVE-2026-88779	Citrix NetScaler	CWE-119			Blocked
2026-10-08	2026-10-11	CVE-2015-3306	ProFTPD	CWE-284			Checked; home study

KEV added	Due	CVE	Product	CWE (KEV)	Ransomware use	Forensic triage	Lab outcome
2026-10-08	2026-10-11	CVE-2015-5477	ISC BIND	CWE-19			Blocked
2026-10-08	2026-10-11	CVE-2016-3081	Apache Struts	CWE-77			Inconclusive
2026-10-08	2026-10-11	CVE-2021-3199	ONLYOFFICE Docs	CWE-22	Unknown	Yes	Checked (Part 2)
2026-10-08	2026-10-11	CVE-2023-22894	Strapi	CWE-312			Blocked
2026-10-13	2026-10-16	CVE-2026-96951	Windows Shell	CWE-77	Unknown	Yes	Checked (Part 2)
2026-10-14	2026-10-17	CVE-2026-105133	AhsayCBS	CWE-287	Unknown	No	Checked (Part 2)
2026-10-14	2026-10-17	CVE-2026-105134	AhsayCBS	CWE-77, CWE-78	Unknown	Yes	Checked (Part 2)
2026-10-15	2026-10-18	CVE-2026-21589	Atlassian	CWE-552			Inconclusive
2026-10-20	2026-10-23	CVE-2026-106931	SiteGuard Server Edition	CWE-78	Unknown	Yes	Checked (Part 2)
2026-10-28	2026-10-31	CVE-2026-46300	Linux kernel	CWE-456	Unknown	Yes	Checked (Part 2)
2026-11-03	2026-11-06	CVE-2026-119977	WordPress	CWE-367	Unknown	Yes	Checked (Part 2)

(**Confirmed**: KEV feed [4]; NVD and JVN list the same CWEs [14] [10] [3].) (Blank cells are fields this study did not record.) Five of the additions are old: on 2026-10-08 CISA listed CVE-2015-3306, CVE-2015-5477, CVE-2016-3081, CVE-2021-3199 and CVE-2023-22894 together, all with the same three-day due date [4]. Every due date in the table except WordPress's had passed by the date of this study.

What the lab could and could not reach

The appliance entries in the window had no lab target: three Ivanti Connect Secure entries, Cisco ASA/FTD, Cisco Secure Email Gateway, Fortinet, PAN-OS, BeyondTrust, Arista VeloCloud and VMware ESX [4]. That is a real limit of a container lab, and it is the part of the list most likely to sit on an organization's edge. The lab ran plans for the 24 entries that had a container image or a local check, plus one for Ghostcat [4] [43]. Of those, this study's checks fell into four groups:

Outcome	Plans
A version verdict on a matched pair or a real host	WordPress, AhsayCBS (two CVEs), ONLYOFFICE (with the JWT precondition), Windows Shell
Read-only host inventory, no version verdict from the plan	Fragnesia (host state recorded; the plan's threshold line did not apply), SiteGuard (product absent)
Blocked: no runnable vulnerable target	BIND, Strapi, NetScaler
Inconclusive: the check ran but could not settle the precondition	Struts, Confluence
Skipped: the check needs a host the lab did not have	Siemens Software Center

(**Demonstrated**, capture files per CVE) [43]

Timeline of the six in-depth flaws

Date	Event	Source
2021-01-26	NVD publishes ONLYOFFICE CVE-2021-3199	[23]
2026-05-13	Fragnesia disclosed on oss-security	[7]
2026-05-23	NVD publishes CVE-2026-46300	[6]
2026-09-29	SiteGuard notifies customers with support contracts	[11]
2026-10-04	NVD publishes the AhsayCBS pair	[14] [16]
2026-10-07	Huntress observes AhsayCBS exploitation from 23:20 UTC (**Reported**)	[18]
2026-10-08	CISA lists CVE-2021-3199	[4]
2026-10-12	Ahsay says its earlier patches were incomplete	[15]
2026-10-13	Microsoft releases CVE-2026-96951, exploited; CISA lists it the same day	[20] [4]
2026-10-14	CISA lists the AhsayCBS pair; Horizon3 publishes its proof of concept repository; SiteGuard and JVN publish CVE-2026-106931	[4] [19] [11] [10]
2026-10-20	CISA lists CVE-2026-106931	[4]
2026-10-26	Microsoft revision 2 adds Edge to CVE-2026-96951	[20]
2026-10-28	WordPress 7.1.4 security release; CISA lists Fragnesia	[2] [4]
2026-10-29	WordPress publishes its advisory GHSA-4w3m-c66w-69mg	[1]
2026-10-31	NVD publishes CVE-2026-119977	[3]
2026-11-03	CISA lists CVE-2026-119977, due 2026-11-06	[4]
2026-11-03 to 2026-11-04	Lab run	[43]

Part 2: six exposure checks in depth

Each section gives the flaw, the public record, why the lab ran a check instead of a test, what the lab showed, and how to find and fix it on real hosts. They run in the order CISA listed them.

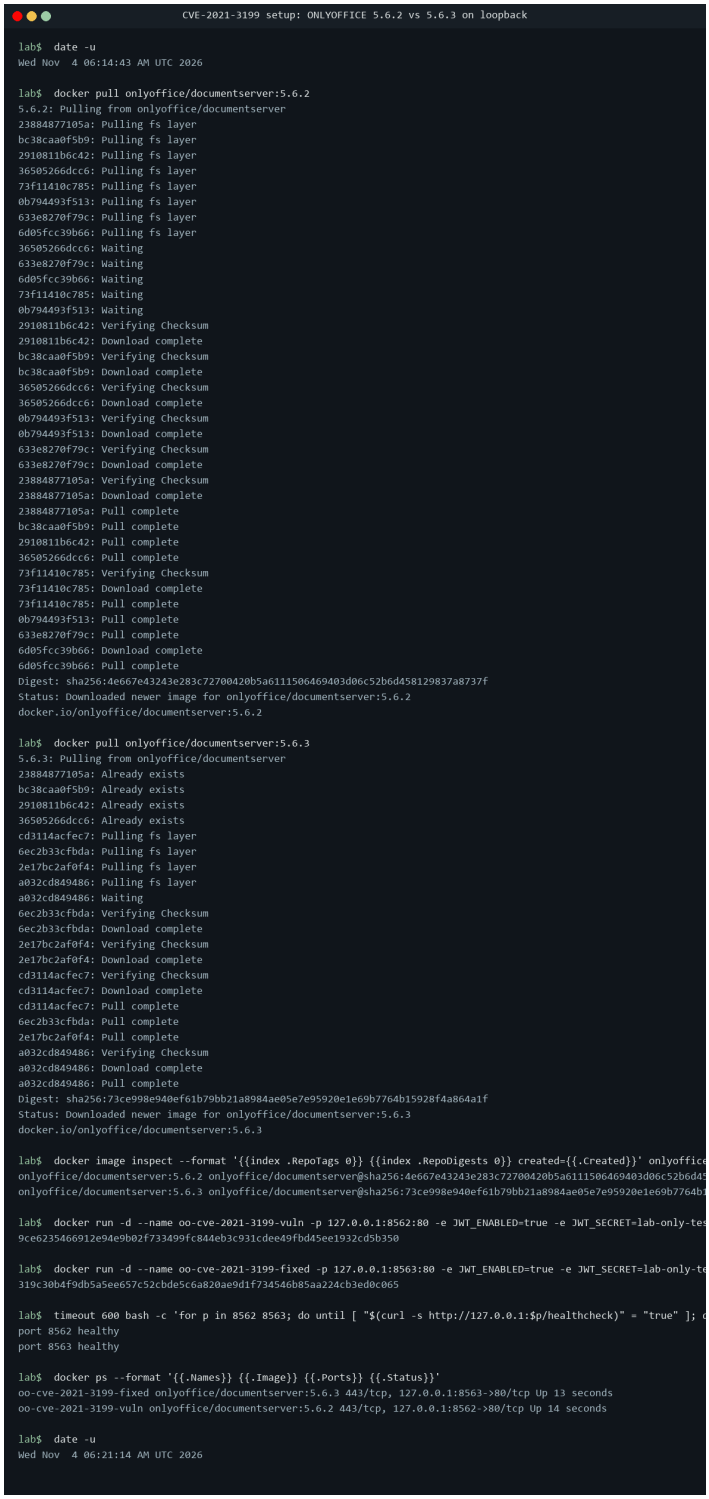
2.1 ONLYOFFICE Document Server, CVE-2021-3199

The flaw. A directory traversal with remote code execution in the `/upload` endpoint of ONLYOFFICE Document Server before 5.6.3, reached through a `/..` sequence in an image upload parameter, and only when JWT is in use (**Confirmed**: NVD) [23]. NVD scores it CVSS 3.1 9.8 Critical with CWE-22 and a CPE of `document_server` below 5.6.3 [23]. The 5.6.3 changelog lists the fix as a path traversal via image upload parameters, Bug #46113 (**Confirmed**) [24].

The record. CISA listed it on 2026-10-08, more than five years after NVD published it, with a due date of 2026-10-11, ransomware use Unknown and forensic triage Yes. Its SSVC entry reads exploitation active, automatable yes, technical impact total (**Confirmed**) [4]. GitHub advisory GHSA-vm8r-r39c-96c6 rates it critical (**Confirmed**) [45].

Why a check. The only public proof of concept, moehw/poc_exploits (created 2021-01-21 and listed as an NVD reference), is classed destructive-or-weaponized in the lab plan, so the lab compared a 5.6.2 container with a 5.6.3 container by `api.js` version, health check, package version and JWT setting, and sent no `/upload` request (**Confirmed**: plan.json) [43].

What the lab showed. Both images were pulled by tag, their digests recorded, and run with JWT enabled, on `127.0.0.1:8562` (5.6.2) and `127.0.0.1:8563` (5.6.3), and both health checks returned `true` (**Demonstrated**, 01-setup.txt) [43].



The plan's first `api.js` regex matched nothing on the minified file; the fallback grep for a quoted version string worked (**Demonstrated**, 02-test.txt) [43]. The package database agreed with it, and the JWT setting was on in both:

```
lab$ for c in oo-cve-2021-3199-vuln oo-cve-2021-3199-fixed; do ... dpkg --compare-versions "$v"
lt 5.6.3 ...; done
```

```
oo-cve-2021-3199-vuln 5.6.2-2 VULNERABLE to CVE-2021-3199 (if JWT enabled)
oo-cve-2021-3199-fixed 5.6.3-2 FIXED for CVE-2021-3199
```

(**Demonstrated**, 02-test.txt; the loop body is shortened here.)

CVE-2021-3199 exposure check

```
lab$  for p in 8562 8563; do echo "port $p:"; curl -s http://127.0.0.1:$p/web-apps/apps/api/documents/api.js | grep
port 8562:
'5.6.2'
port 8563:
'5.6.3'

lab$  for p in 8562 8563; do echo "port $p healthcheck: $(curl -s http://127.0.0.1:$p/healthcheck)"; done
port 8562 healthcheck: true
port 8563 healthcheck: true

lab$  for c in oo-cve-2021-3199-vuln oo-cve-2021-3199-fixed; do echo "$c:"; docker exec $c sh -c "dpkg-query -W -f='
oo-cve-2021-3199-vuln:
onlyoffice-documentserver 5.6.2-2
oo-cve-2021-3199-fixed:
onlyoffice-documentserver 5.6.3-2

lab$  rpm -q onlyoffice-documentserver onlyoffice-documentserver-ee 2>/dev/null || echo "rpm query: no RHEL/CentOS p
package onlyoffice-documentserver is not installed
package onlyoffice-documentserver-ee is not installed
rpm query: no RHEL/CentOS package (Debian-based lab host and containers)

lab$  docker ps --format '{{.Names}} {{.Image}}' | grep -i onlyoffice/documentserver
oo-cve-2021-3199-fixed onlyoffice/documentserver:5.6.3
oo-cve-2021-3199-vuln onlyoffice/documentserver:5.6.2

lab$  for c in oo-cve-2021-3199-vuln oo-cve-2021-3199-fixed; do echo "$c:"; docker exec $c grep -n -A6 '"token"' /et
oo-cve-2021-3199-vuln:
12:      "token": {
13-        "enable": {
14-          "request": {
15-            "inbox": true,
16-            "outbox": true
17-          },
18-          "browser": true
oo-cve-2021-3199-fixed:
12:      "token": {
13-        "enable": {
14-          "request": {
15-            "inbox": true,
16-            "outbox": true
17-          },
18-          "browser": true
```

The mitigation check confirmed the fixed container runs 5.6.3-2, that the JWT secrets on both are set and not the default, and that both listen on loopback only (**Demonstrated**, 04-mitigation.txt) [43].

CVE-2021-3199 mitigation verification (read-only)

```
lab$  docker ps --filter name=oo-cve-2021-3199 --format '{{.Names}} image={{.Image}}'
oo-cve-2021-3199-fixed image=onlyoffice/documentserver:5.6.3
oo-cve-2021-3199-vuln image=onlyoffice/documentserver:5.6.2

lab$  for c in oo-cve-2021-3199-vuln oo-cve-2021-3199-fixed; do echo "$c: $(docker exec $c dpkg-query -W -f='${Versi
oo-cve-2021-3199-vuln: 5.6.2-2
oo-cve-2021-3199-fixed: 5.6.3-2

lab$  for c in oo-cve-2021-3199-vuln oo-cve-2021-3199-fixed; do echo "$c:"; docker exec $c python3 -c "import json;c
oo-cve-2021-3199-vuln:
  inbox/outbox/session secret set and non-default = True
oo-cve-2021-3199-fixed:
  inbox/outbox/session secret set and non-default = True

lab$  for c in oo-cve-2021-3199-vuln oo-cve-2021-3199-fixed; do echo "$c: $(docker port $c)"; done
oo-cve-2021-3199-vuln: 80/tcp -> 127.0.0.1:8562
oo-cve-2021-3199-fixed: 80/tcp -> 127.0.0.1:8563

lab$  ss -ltn | grep -E ':(8562|8563)\b'
LISTEN 0      4096       127.0.0.1:8562       0.0.0.0:*
LISTEN 0      4096       127.0.0.1:8563       0.0.0.0:*
```

Cleanup removed both containers, their 14 volumes and both images, and the ports were clear (**Demonstrated**, 05-cleanup.txt) [43].

CVE-2021-3199 cleanup

```
lab$  for c in oo-cve-2021-3199-vuln oo-cve-2021-3199-fixed; do docker inspect --format '{{range .Mounts}}{{if eq .T
14

lab$  docker rm -f oo-cve-2021-3199-vuln oo-cve-2021-3199-fixed
oo-cve-2021-3199-vuln
oo-cve-2021-3199-fixed

lab$  xargs -r docker volume rm < /tmp/lab/CVE-2021-3199/volumes.txt | wc -l
14

lab$  docker rmi onlyoffice/documentserver:5.6.2 onlyoffice/documentserver:5.6.3
Untagged: onlyoffice/documentserver:5.6.2
Untagged: onlyoffice/documentserver@sha256:4e667e43243e283c72700420b5a6111506469403d06c52b6d458129837a8737f
Deleted: sha256:e7f53e4bfa86081e15d0e2bb6e901b091a658237064af221058a44151f4fea60
Deleted: sha256:2212ec9cd5b959893d6f85a37c4b00bb09dcf1310e10d52fab375cb5290b5dcd
Deleted: sha256:e6c00118e241e6f35659031b29e39824df4ee0f7e5c087fb34223f255d21d6de
Deleted: sha256:c4d693e57fb80bca01f631a39e82f52a9d64b47ef9d086c23f8aa4101e008f31
Deleted: sha256:747f20a7d375e1291d64b5e05b961916e3793f04ab4c851afa57994efd8029de
Untagged: onlyoffice/documentserver:5.6.3
Untagged: onlyoffice/documentserver@sha256:73ce998e940ef61b79bb21a8984ae05e7e95920e1e69b7764b15928f4a864a1f
Deleted: sha256:b70e4b3d35b7938fbb744714cced451b14c6950942721bf4bb8d9b548f2d91e1
Deleted: sha256:0fbda5cc33423d44db77a759205f42bc670a1d5c438636c61b713fc5f9dad764
Deleted: sha256:4606ebb9dd722f8e600d5edd50215f4ae58b7a425e6b1686e42713686a87ff11
Deleted: sha256:227a8c3a717753e747afe67a615949e67ead622ddf38d3f09e4beca4d5da6174
Deleted: sha256:bae05d07243d595eb44cfd9c6a981a5b3ed7217a704d4e0e36dd13528a72a491
Deleted: sha256:82751c4dcc36265daceeee0abd938806b591eb66700863186377348c96e8aead
Deleted: sha256:24b16cac34768092d57cdb429a12de6c9ee870a3724d70098a189ae0ac344c42
Deleted: sha256:b1daff3b271fede2bec31812c398e28952f792935f608e467271cdfaccab7de2
Deleted: sha256:b7f7d2967507ba709dbd1dd0426a5b0cdbe1ff936c131f8958c8d0f910eea19e

lab$  ss -ltnp | grep -E ':(8562|8563)\b' || echo 'ports 8562/8563 clear'
ports 8562/8563 clear

lab$  docker ps -a --format '{{.Names}} {{.Image}}'; echo "containers listed above (none of oo-cve-2021-3199-*)"
containers listed above (none of oo-cve-2021-3199-*)

lab$  docker images --format '{{.Repository}}:{{.Tag}}' | grep -i onlyoffice || echo 'no onlyoffice images'
no onlyoffice images

lab$  docker volume ls -q | grep -cFf /tmp/lab/CVE-2021-3199/volumes.txt || true
0

lab$  rm -rf /tmp/lab/CVE-2021-3199/

lab$  ls -d /tmp/lab/CVE-2021-3199
ls: cannot access '/tmp/lab/CVE-2021-3199': No such file or directory

lab$  docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

Find and fix it. On Debian-family installs, `dpkg-query -W -f='${Package} ${Version}\n' onlyoffice-documentserver` and compare with 5.6.3; on RPM installs, `rpm -q onlyoffice-documentserver`; from the network, the quoted version string in `/web-apps/apps/api/documents/api.js` (**Demonstrated** method) [43]. Then read the `token.enable` block in `/etc/onlyoffice/documentserver/local.json`. The NVD text makes JWT a condition of the flaw [23], so

an old server with JWT on is the exposed case; that turning on a hardening feature is what opens the path is this study's reading (**Inferred**). Upgrade to 5.6.3 or later [24] [23].

2.2 Windows Shell, CVE-2026-96951

The flaw. A Windows Shell security feature bypass, CWE-77, released on 2026-10-13. A successful exploit evades Mark of the Web: the attacker sends a crafted file by email or instant message, and the user has to click a link or open the file (**Confirmed**: Microsoft) [20]. Microsoft rates it Important with CVSS 8.8 base and 8.2 temporal (`E:F/RL:O/RC:C`), marks it exploited and not publicly disclosed, and added Edge in revision 2 on 2026-10-26 [20].

The record. NVD (Analyzed) sets fixed-build thresholds per Windows release; any build below its line is vulnerable (**Confirmed**) [21]:

Release	Fixed build (NVD threshold)
Windows 10 1607 and Server 2016	10.0.14393.9612
Windows 10 1809 and Server 2019	10.0.17763.9376
Windows 10 22H2	10.0.19045.7807
Windows 11 23H2	10.0.22631.7652
Windows 11 24H2	10.0.26100.9711
Windows 11 25H2	10.0.26200.9711
Windows 11 26H2	10.0.26300.9711
Server 2022	10.0.20348.5761

CISA listed it on 2026-10-13, due 2026-10-16, ransomware use Unknown, forensic triage Yes; SSVC active, not automatable, total (**Confirmed**) [4]. A GitHub repository search for the CVE returned no results on 2026-11-04 [22]. Microsoft's `E:F` temporal metric says it judges functional exploit code to exist, which is not the same as public code (**Inferred**) [20] [22].

Why a check. No proof of concept was found and the flaw is in an operating-system component with no container image, so the lab read the build number and installed updates on a Windows workstation and compared them with the fixed build (**Confirmed**: plan.json) [43].

What the lab showed. The plan created nothing to set up: no images, no containers, and no containers from other plans were running on the lab host (**Demonstrated**, 01-setup.txt) [43].

```
CVE-2026-96951 - setup (none; preflight)

lab$  echo 'CVE-2026-96951 exposure-check: plan setup list is empty, no images or containers'
CVE-2026-96951 exposure-check: plan setup list is empty, no images or containers

lab$  ssh lab-host docker ps --format '{{.Names}}'   (preflight: no containers from other plans)
(no running containers)
```

The workstation (hostname redacted to `ASSESSED-PC`) reported build 26300 with an update build revision of `0x265a`, which is 9818, so build 26300.9818. Security Update KB5127647 was installed on 10/16/2026, and Edge was 156.0.4314.58 (**Demonstrated**, 03-exposure.txt) [43].

```
CVE-2026-96951 - exposure check

lab$  reg query "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion" /v CurrentBuild

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion
    CurrentBuild    REG_SZ    26300

lab$  reg query "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion" /v UBR

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion
    UBR    REG_DWORD    0x265a

lab$  powershell -NoProfile -Command "$cv = Get-ItemProperty 'HKLM:\SOFTWARE\Microsoft\Windows NT\CurrentVersion'; '
26300.9818

lab$  powershell -NoProfile -Command "Get-HotFix | Where-Object { $_.InstalledOn -ge [datetime]'2026-10-13' }"

Source        Description      HotFixID      InstalledBy          InstalledOn
------        -----------      --------      -----------          -----------
ASSESSED-PC      Update           KB5127694     NT AUTHORITY\SYSTEM  10/18/2026 12:00:00 AM
ASSESSED-PC      Update           KB5121794     NT AUTHORITY\SYSTEM  10/18/2026 12:00:00 AM
ASSESSED-PC      Update           KB5127936     NT AUTHORITY\SYSTEM  11/3/2026 12:00:00 AM
ASSESSED-PC      Security Update  KB5127647     NT AUTHORITY\SYSTEM  10/16/2026 12:00:00 AM
ASSESSED-PC      Update           KB5132402     NT AUTHORITY\SYSTEM  11/1/2026 12:00:00 AM

lab$  powershell -NoProfile -Command "(Get-Item \"${env:ProgramFiles(x86)}\Microsoft\Edge\Application\msedge.exe\").
156.0.4314.58
```

Both version comparisons came back `True`: the build against 10.0.26300.9711 and Edge against the plan's Edge threshold of 154.0.4258.62 (**Demonstrated**, 04-mitigation.txt) [43].

!Terminal output on a Windows workstation: a PowerShell [version] comparison of 10.0 plus CurrentBuild and UBR against 10.0.26300.9711 prints True, and a [version] comparison of the Edge product version against 154.0.4258.62 prints True

```
CVE-2026-96951 - cleanup

lab$  echo 'cleanup: plan created no containers, networks or images; no TOKEN/WAZUH_* variables were set (Wazuh test
cleanup: plan created no containers, networks or images; no TOKEN/WAZUH_* variables were set (Wazuh tests skipped)

lab$  ssh lab-host docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

lab$  ssh lab-host ls -d /tmp/lab/CVE-2026-96951   (expect: not present)
ls: cannot access '/tmp/lab/CVE-2026-96951': No such file or directory
not present
```

(**Demonstrated**, 05-cleanup.txt) [43]

Find and fix it. Read `CurrentBuild` and `UBR` from `HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion`, build `[version]('10.0.{0}.{1}' -f $cv.CurrentBuild, $cv.UBR)` in PowerShell, and compare it with the threshold for that release in the table above; `Get-HotFix` shows the update that moved it (**Demonstrated** method) [43] [21]. Compare as a version, not as text: a text comparison misorders revision numbers of different lengths (**Inferred**). Check Edge too, since Microsoft added it in revision 2 [20].

2.3 AhsayCBS, CVE-2026-105133 and CVE-2026-105134

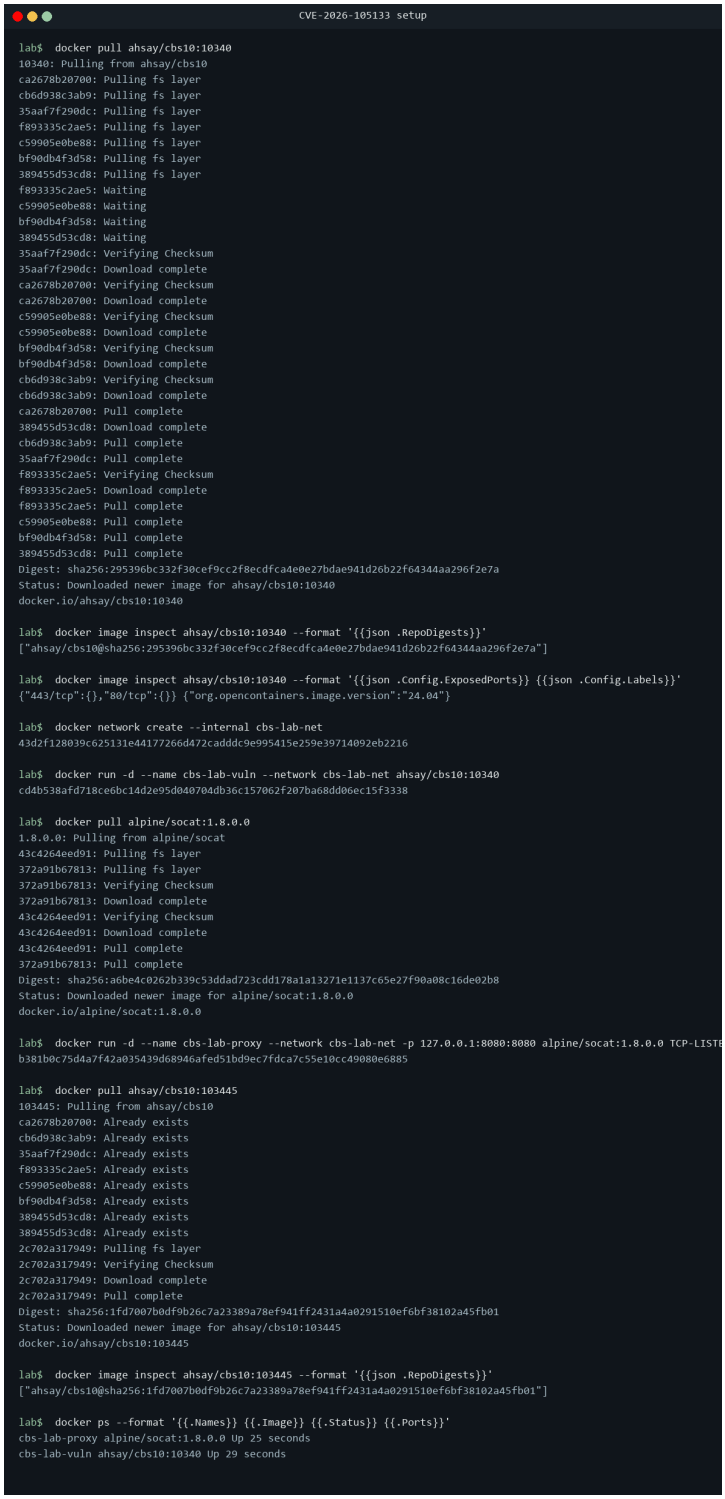
The flaws. CVE-2026-105133 is improper authentication in `checkSysPwd` (`com/ahsay/obs/api/ApiStructsAction.java`) of the AhsayCBS backup server API, reachable remotely through the `random` argument. It affects up to 9.15.2.69 and 10.3.4.44 and is fixed in 9.15.2.70 and 10.3.4.45; NVD notes the initial patch did not fix it properly (**Confirmed**) [14]. The CNA, VulDB, scores it CVSS 3.1 7.3 High and CVSS 4.0 6.9 Medium, CWE-287 [14]. CVE-2026-105134 is described by NVD and KEV as OS command injection in the Replication Receiver at `/rps/api/json/UpdateReceivers.do` through the `random` argument, remote and unauthenticated, with the same affected and fixed versions, CVSS 3.1 10.0 Critical (scope changed) and CVSS 4.0 10.0, CWE-77 and CWE-78 (**Confirmed**) [16].

A disagreement on mechanism. Horizon3 describes the exploitable CVE-2026-105134 step as a path traversal in the replication upload that writes a JSP file, with `UpdateReceivers.do` only configuring the receiver, which differs from the NVD and KEV wording (**Reported**; opened 2026-11-03, not re-opened) [17]. The fixed version is the same either way.

The record. Ahsay's notice of October 12, 2026 says the earlier patches, v9.15.2.59 and v10.3.4.0, did not fully address either CVE, and tells customers to apply hotfix v10.3.4.45 (v10) or v9.15.2.70 (v7, v8 and v9) from the partner portal and then reboot (**Confirmed**) [15]. CISA listed both on 2026-10-14, due 2026-10-17, ransomware use Unknown, as a chain. The two entries differ: CVE-2026-105133 has forensic triage No and SSVC active, automatable yes, partial; CVE-2026-105134 has forensic triage Yes and SSVC active, automatable yes, total (**Confirmed**) [4]. Huntress observed exploitation of the chain from 2026-10-07 23:20 UTC, five organisations by 2026-10-08, with JSP web shells and XMRig miners (**Reported**; opened 2026-11-03, not re-opened) [18].

Why a check. A public proof of concept, horizon3ai/AhsayCBS-Auth-Bypass-RCE (created 2026-10-14, no licence), exists [19]. The lab plan classes it destructive-or-weaponized because it creates an admin account or plants a web shell, so it was recorded only, and the lab compared the tags, digests and installed versions of throwaway `10340` and `103445` containers (**Confirmed**: GitHub API and plan.json) [19] [43].

What the lab showed. The vulnerable image was pulled by tag, its digest recorded, and the container started on an internal Docker network behind a loopback-only proxy (**Demonstrated**, CVE-2026-105133 01-setup.txt) [43].



The install metadata settled it. `/usr/local/cbs/version.txt` read `10.3.4.0` in the `ahsay/cbs10:10340` container and `10.3.4.45` in the `103445` container, and `docker images --digests` tied each tag to its digest (**Demonstrated**, CVE-2026-105134 03-exposure.txt) [43]. The lab host itself had no `/usr/local/cbs`. The Windows checks in the plan (the `cbssvc` service, the uninstall registry and a listing of `.jsp` files under the web apps folder) do not apply on a Linux Docker host and were not run.

CVE-2026-105134 AhsayCBS exposure check - version files, images, containers

```
# CVE-2026-105134 exposure_check commands (2026-11-04T06:12:08Z)
# Linux check, run inside each container (the AhsayCBS install lives there, not on the Docker host)
lab$  docker exec cbs-vuln sh -c 'find / -xdev -maxdepth 5 -iname "version*.txt" -path "*cbs*" 2>/dev/null -exec sh
== /usr/local/cbs/version.txt
10.3.4.0
lab$  docker exec cbs-fixed sh -c 'find / -xdev -maxdepth 5 -iname "version*.txt" -path "*cbs*" 2>/dev/null -exec sh
== /usr/local/cbs/version.txt
10.3.4.45

# Same Linux check on the Docker host itself (no native install expected)
lab$  find / -xdev -maxdepth 5 -iname 'version*.txt' -path '*cbs*' 2>/dev/null -exec sh -c 'echo "== $1"; head -n 3
host find exit=1

# Docker host: AhsayCBS images and digests (vulnerable 10340 digest = sha256:295396bc332f30cef9cc2f8ecdfca4e0e27bdae
lab$  docker images --digests --no-trunc | grep -Ei 'ahsay|cbs'
ahsay/cbs10   103445    sha256:1fd7007b0df9b26c7a23389a78ef941ff2431a4a0291510ef6bf38102a45fb01   sha256:bdbc7ecd2a0
ahsay/cbs10   10340     sha256:295396bc332f30cef9cc2f8ecdfca4e0e27bdae941d26b22f64344aa296f2e7a   sha256:857fc52276c

# Docker host: running containers on an AhsayCBS image. This plan's containers are cbs-vuln, cbs-fixed and cbs-vuln-
lab$  docker ps --format '{{.Names}} {{.Image}}' | grep -Ei 'ahsay|cbs'
cbs-fixed ahsay/cbs10:103445
cbs-vuln ahsay/cbs10:10340
cbs-vuln-web ahsay/cbs10:10340

# Windows (PowerShell) checks in the plan (Win32_Service cbssvc, Uninstall registry, webapps .jsp listing): not appl
# Decision rule: v7 to v9 affected below 9.15.2.70; v10 affected below 10.3.4.45. 10.3.4.0 = AFFECTED, 10.3.4.45 = F
```

```
lab$ for c in cbs-vuln cbs-fixed; do ... sort -V ...; done
cbs-vuln 10.3.4.0 AFFECTED
cbs-fixed 10.3.4.45 FIXED
```

(**Demonstrated**, CVE-2026-105134 04-mitigation.txt; the loop body is shortened here.) No request was sent to any `/rps/` or `.do` endpoint, and the proxy block, the post-patch hunt and the reboot were not exercised (04-mitigation.txt).

CVE-2026-105134 AhsayCBS mitigation verification - affected vs fixed build

```
# CVE-2026-105134 mitigation verification, read-only (2026-11-04T06:12:10Z)
# Mitigation = vendor hotfix 10.3.4.45 (v10) or 9.15.2.70 (v7-v9). 10.3.4.0 is NOT a fix. The check below confirms w
lab$  docker exec cbs-vuln cat /usr/local/cbs/version.txt; docker exec cbs-fixed cat /usr/local/cbs/version.txt
10.3.4.010.3.4.45

lab$  for c in cbs-vuln cbs-fixed; do v=$(docker exec $c cat /usr/local/cbs/version.txt | tr -d '\r\n '); if [ "$(pr
cbs-vuln 10.3.4.0 AFFECTED
cbs-fixed 10.3.4.45 FIXED

# Not exercised in the lab: WAF/proxy block of /rps/, post-patch compromise hunt, reboot. No request was sent to any
```

The 10.3.4.0 reading matters: it is one of the two builds Ahsay itself calls an incomplete patch [15]. A host that applied the first fix and stopped looks patched to anyone who checks only "did we apply the October hotfix" (**Inferred**). Cleanup removed all three containers, the plan's own internal network and both images, after checking that no other container still used them (**Demonstrated**, CVE-2026-105134 05-cleanup.txt) [43].

CVE-2026-105134 AhsayCBS cleanup - containers, network, images removed

```
# CVE-2026-105134 cleanup (2026-11-04T06:12:34Z)
lab$  docker rm -f cbs-vuln cbs-fixed cbs-vuln-web
cbs-vuln
cbs-fixed
cbs-vuln-web

# This plan's network is cbs-105134-net (see setup deviation); cbs-lab-net belonged to the CVE-2026-105133 plan and
lab$  docker network rm cbs-105134-net
cbs-105134-net

# Confirm no other container still uses the AhsayCBS images before removing them
lab$  docker ps -a --format '{{.Names}} {{.Image}}' | grep -i ahsay || echo 'no container uses an ahsay image'
no container uses an ahsay image

lab$  docker rmi ahsay/cbs10:10340 ahsay/cbs10:103445
Untagged: ahsay/cbs10:10340
Untagged: ahsay/cbs10@sha256:295396bc332f30cef9cc2f8ecdfca4e0e27bdae941d26b22f64344aa296f2e7a
Deleted: sha256:857fc52276cbfd02f7fff98dab90ef54bccf5b8bb9922a8a926d21413ac5c735
Untagged: ahsay/cbs10:103445
Untagged: ahsay/cbs10@sha256:1fd7007b0df9b26c7a23389a78ef941ff2431a4a0291510ef6bf38102a45fb01
Deleted: sha256:bdbc7ecd2a0bf9787d531b1721dceb98cd4aa48854082a59f8a5a456698b0069
Deleted: sha256:0e2921a3d878c6dffdcd7da8b983962b47f26c263c513e0fd15fe5dc2cc8e61f
Deleted: sha256:e2fcf1e84c351f4faf577e688e8d0534948ebd79863d6b5bca6ac38e87f5f818
Deleted: sha256:29cad1bba6c651d45e8f0b8a43326075c0a22eec22bbd2998fcd09ebe0bf3042
Deleted: sha256:da6a61dba2237617f60f2210f6b2934913c5f54e3df071fbc140670665abca3e
Deleted: sha256:33b18e4da3717c6be7a5e9ee0d33fa22cb9f783029de2798012a85a78e3b5985
Deleted: sha256:b6ec280df99970d57637ea1e1789159561137c541065dd7242bb65e7f33fff52
Deleted: sha256:3d5daa77f67f07276f6189eb0708493f5789ddd9994f3f9e9d4a83ebb62a0bbf
Deleted: sha256:244c7533c841d8590fc77038eb28424c24ec646034edb31de92e2b9aa6380824
Deleted: sha256:f103cd120fdd6cbc7f3d0e1d5bfb3e16290c32ead7b822d15c34460ecdf64b29

# DEVIATION: the plan's global 'docker volume prune -f; docker image prune -f' is replaced by targeted checks, so ot
lab$  docker images --filter dangling=true --format '{{.ID}} {{.Size}}'; echo 'dangling images listed above (none ex
dangling images listed above (none expected)

lab$  docker ps -a --filter name=cbs-; docker images | grep -i ahsay || echo 'clean'
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
clean

lab$  docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

lab$  docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
c6196556e799   bridge    bridge    local
c41a82569086   host      host      local
c1d6680892f7   none      null      local

lab$  rm -rf /tmp/lab/CVE-2026-105134/

lab$  ls /tmp/lab/
CVE-2026-96655

lab$  docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

Find and fix it. On Linux installs read `/usr/local/cbs/version.txt`; on Windows the plan's checks were the `cbssvc` service and the uninstall registry entry, which the lab did not run [43]. Treat 10.3.4.0 and 9.15.2.59 as vulnerable, and upgrade to 10.3.4.45 (v10) or 9.15.2.70 (v7 to v9), then reboot [15]. Given Huntress's account of JSP web shells and miners [18], look for `.jsp` files that do not belong to the product and for

unexpected miner processes before you close the ticket (**Inferred** from a **Reported** source).

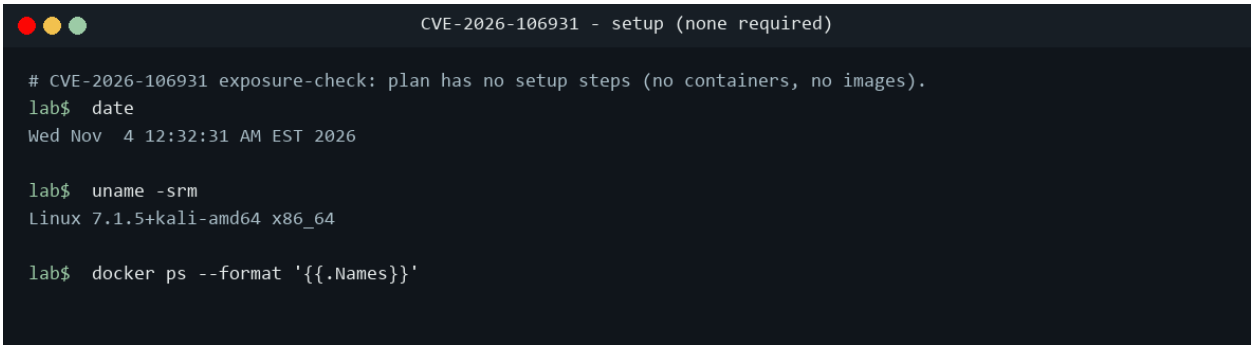
2.4 SiteGuard Server Edition, CVE-2026-106931

The flaw. OS command injection, CWE-78, in the statistics feature of SiteGuard Server Edition, a host-based web application firewall. An attacker may run arbitrary OS commands as root or SYSTEM (**Confirmed**: JVN) [10].

The record. Affected are Ver2.20 and later with the statistics feature enabled, which is on by default: the Apache, Nginx and IIS editions before Ver8.00 Update5, and the FreeBSD Apache and Nginx editions before Ver7.00 Update6. IIS Ver3.10 to Ver4.00 Update4 are not affected. The vendor offers a file-replacement patch, including for end-of-life versions (**Confirmed**: vendor) [11]. The vendor tells customers to download the diagnostic information before updating, because the update or patch may erase attack traces, and says customers with support contracts were notified on 2026-09-29 [11]. JPCERT/CC scores it CVSS 4.0 8.5 and CVSS 3.0 7.8 with `AV:L/PR:L`; the vendor and JVN both state exploitation in the wild is confirmed; NVD published it on 2026-10-14 and lists it as Analyzed (**Confirmed**) [12] [10] [11]. CISA listed it on 2026-10-20, due 2026-10-23, ransomware use Unknown, forensic triage Yes; SSVC active, not automatable, total [4]. A GitHub repository search for the CVE returned no results [13].

Why a check. No proof of concept was found and the product is commercial and licensed with no official image, so the lab ran a read-only package, configuration and process inventory on a host without SiteGuard. The result is a negative one (**Confirmed**: plan.json) [43].

What the lab showed. There was nothing to set up (**Demonstrated**, 01-setup.txt) [43].



Every check came back empty: `dpkg`, `rpm`, `pkg`, the configuration grep across the Apache and Nginx folders, and the process listing, ending in `siteguard package matches: 0` (**Demonstrated**, 03-exposure.txt) [43].

CVE-2026-106931 - exposure check

```
# Exposure check list from the plan (Linux items; Windows/IIS and FreeBSD items not applicable to this Kali host).
lab$  dpkg -l 2>/dev/null | grep -i siteguard

lab$  rpm -qa 2>/dev/null | grep -i siteguard

lab$  pkg info 2>/dev/null | grep -i siteguard

lab$  grep -RIn -i siteguard /etc/httpd /etc/apache2 /etc/nginx /usr/local/etc 2>/dev/null

lab$  ps -eo user,pid,ppid,cmd | grep -i [s]iteguard

lab$  echo "siteguard package matches: $( (dpkg -l 2>/dev/null; rpm -qa 2>/dev/null) | grep -ci siteguard )"
siteguard package matches: 0
```

One step gave a false positive worth knowing about. `pgrep -f -i siteguard` matched the capture wrapper's own shell, because the wrapper's command line contained the word, and the only child of that "parent" was the `ps` command doing the checking (**Demonstrated**, 02-test.txt) [43]. A process-name check that searches full command lines will find itself; filter on the executable name, or use the bracket trick (`grep -i [s]iteguard`) the plan used for `ps` (**Inferred**).

CVE-2026-106931 - mitigation (not applicable)

```
# Mitigation steps (vendor update to Ver8.00 Update5 / Ver7.00 Update6, statistics feature off) are changes on a lic
# SiteGuard is absent on this host, so there is nothing to patch and no read-only mitigation check applies. Read-onl
lab$  dpkg -l 2>/dev/null | grep -i siteguard

lab$  ps -eo user,pid,ppid,cmd | grep -i [s]iteguard
```

CVE-2026-106931 - cleanup

```
# Plan created no containers, networks or images. auditd rules were never added (no sudo), so auditctl removal is no
lab$  rm -f /tmp/siteguard-exposure-$(hostname).txt; ls -l /tmp/siteguard-exposure-* 2>&1
ls: cannot access '/tmp/siteguard-exposure-*': No such file or directory

lab$  docker ps -a --format '{{.Names}} {{.Image}} {{.Status}}'

lab$  docker network ls --format '{{.Name}}'
bridge
host
none

lab$  rm -rf /tmp/lab/CVE-2026-106931/
lab$  ls -ld /tmp/lab/CVE-2026-106931
ls: cannot access '/tmp/lab/CVE-2026-106931': No such file or directory
lab$  docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

(**Demonstrated**, 04-mitigation.txt and 05-cleanup.txt) [43]

Find and fix it. On Linux, `dpkg -l | grep -i siteguard` and `rpm -qa | grep -i siteguard`; on FreeBSD, `pkg info | grep -i siteguard`; then grep the web server configuration for SiteGuard modules (**Demonstrated** method on Linux) [43]. The admin console's version and the statistics-feature setting were not checked, because no SiteGuard console existed on the lab host (02-test.txt). On a host that has it: download the diagnostic information first, then update to Ver8.00 Update5 (Apache, Nginx, IIS) or Ver7.00 Update6 (FreeBSD), or apply the vendor's file-replacement patch [11].

2.5 Linux kernel, CVE-2026-46300 (Fragnesia)

The flaw. `skb_try_coalesce()` loses the `SKBFL_SHARED_FRAG` marker when it moves paged fragments, so ESP input can decrypt in place over fragments backed by the page cache; the fix propagates the marker (**Confirmed**: NVD) [6]. It was disclosed on oss-security on 2026-05-13 as a Dirty Frag class local privilege escalation in XFRM ESP-in-TCP, discovered by William Bowling with the V12 team; no race is required, and the mitigation is the same as for Dirty Frag (**Confirmed**) [7]. Red Hat says a low-privileged local attacker can write into the page cache of read-only files and overwrite privileged binaries to gain root, rates it Important because access is local, and gives the mitigation in bulletin RHSB-2026-003 [8].

The record. NVD scores it CVSS 3.1 7.8 High (`AV:L/AC:L/PR:L/UI:N`) and lists CWE-787 (NVD), CWE-456 (CISA ADP) and CWE-123 [6]. Upstream affected ranges are 3.9 through 5.10.257, 5.11 to before 5.15.208, 5.16 to before 6.1.174, 6.2 to before 6.6.141, 6.7 to before 6.12.91, 6.13 to before 6.18.33, 6.19 to before 7.0.10 and the 7.1 release candidates, plus Siemens SIMATIC S7-1500 CPU 1518-4 PN/DP MFP firmware 3.1.5 to 3.1.6 (**Confirmed**) [6]. Ubuntu rates it high and lists fixed kernels of 5.15.0-181.191 (22.04), 6.8.0-124.124 (24.04), 6.17.0-35.35 (25.10) and 7.0.0-22.22 (26.04); 14.04, 16.04, 18.04 and 20.04 are not affected [9]. CISA listed it on 2026-10-28, due 2026-10-31, ransomware use Unknown, forensic triage Yes, CWE-456, and states it can be chained with CVE-2026-109618 and CVE-2026-109619 to implant PHP web shells and run commands as root; SSVC active, not automatable, total (**Confirmed**) [4]. The PHP half of that chain is the subject of the PHP2Shell study and of the PHP 8.4.27 lab study.

Why a check. The plan classes it version-check. A kernel flaw is shared by every container on the host, and every public trigger rewrites the host's page cache of a real setuid binary, so the lab compared kernel package versions and module and user-namespace state only (**Inferred**: the plan's reasoning) [43].

What the lab showed. No container applies; the check ran read-only on the lab host (**Demonstrated**, 01-setup.txt) [43].

CVE-2026-46300 setup (no container)

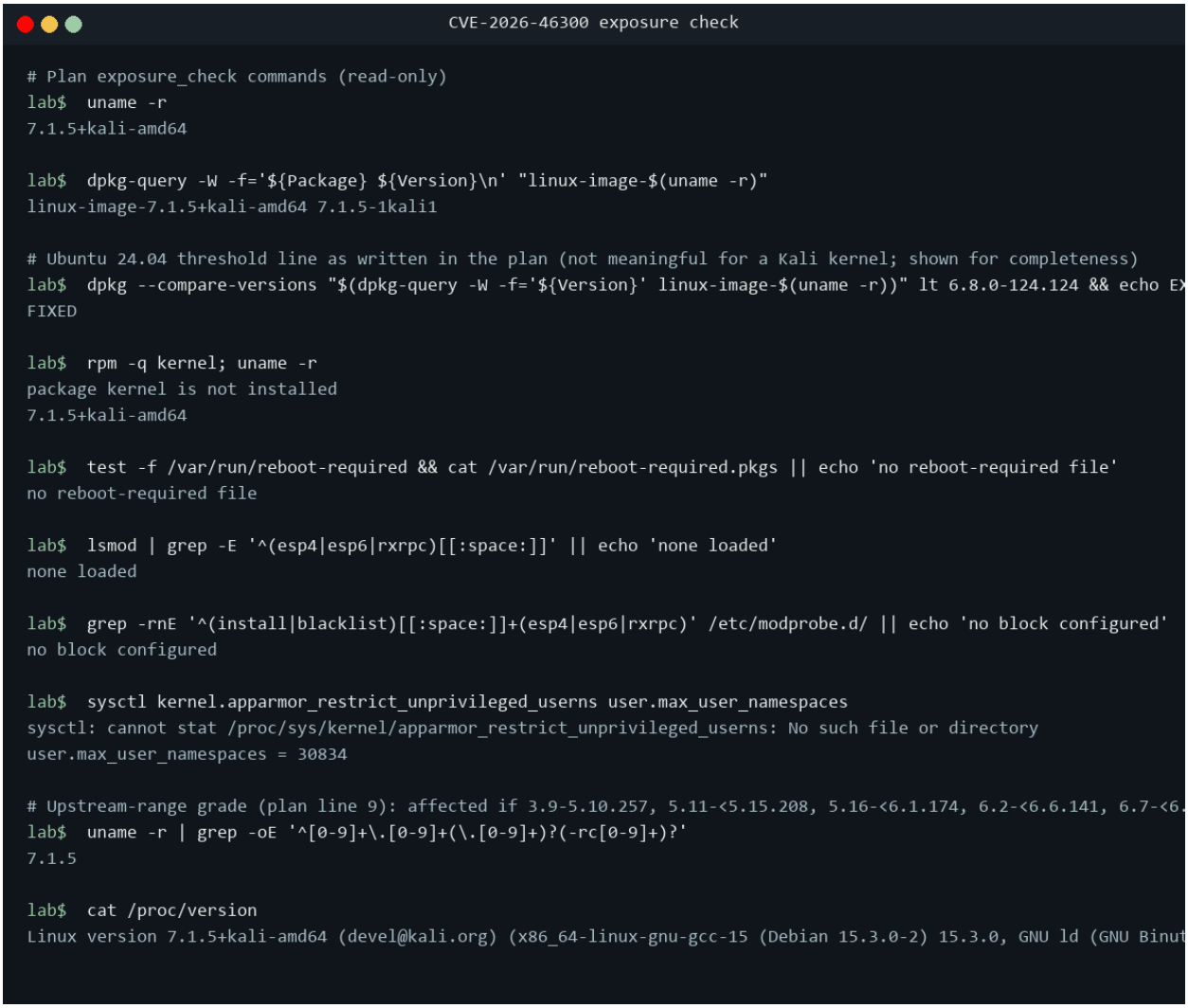
```
# CVE-2026-46300 (Fragnesia) exposure check. Plan setup list is empty: kernel bug, no container applies.
lab$  mkdir -p /tmp/lab/CVE-2026-46300

lab$  hostname; date -u
lab-host
Wed Nov  4 05:28:36 AM UTC 2026

lab$  docker ps --format '{{.Names}}'
```

The running kernel was `7.1.5+kali-amd64` from package `7.1.5-1kali1`, and a second image, `linux-image-6.19.14+kali-amd64` at `6.19.14-1+kali1`, was also installed. `xfrm_user` was loaded; `esp4`, `esp6` and `rxrpc` were not loaded but were present and loadable, with no block in `/etc/modprobe.d/`; and

`kernel.unprivileged_userns_clone` was `1` (**Demonstrated**, 02-test.txt, 03-exposure.txt and 04-mitigation.txt) [43].



Two things in that output need care. First, the plan's grading line compared the Kali package version with Ubuntu 24.04's fixed build and printed `FIXED`, which means nothing for a Kali kernel; the version strings of two distributions do not order against each other (**Demonstrated**, 02-test.txt; the reading is **Inferred**). Second, read against NVD's upstream ranges, the running 7.1.5 is a 7.1 stable release, outside the listed ranges, while the fallback 6.19.14 image falls in "6.19 to before 7.0.10" (**Inferred** from [6]). A host that boots its fallback kernel after a failed update would boot into an affected one (**Inferred**).

The plan also took a forensic baseline. The SHA-256 of `/usr/bin/su` read through the page cache and read with `O_DIRECT` from disk matched, and the `O_DIRECT` MD5 matched the `util-linux` record in the dpkg database (**Demonstrated**, 02-test.txt) [43]. Because Fragnesia writes into the page cache and not the file on disk [8], a mismatch between the cached and direct reads of a setuid binary would be the sign to look for (**Inferred**).

CVE-2026-46300 mitigation verification (read-only)

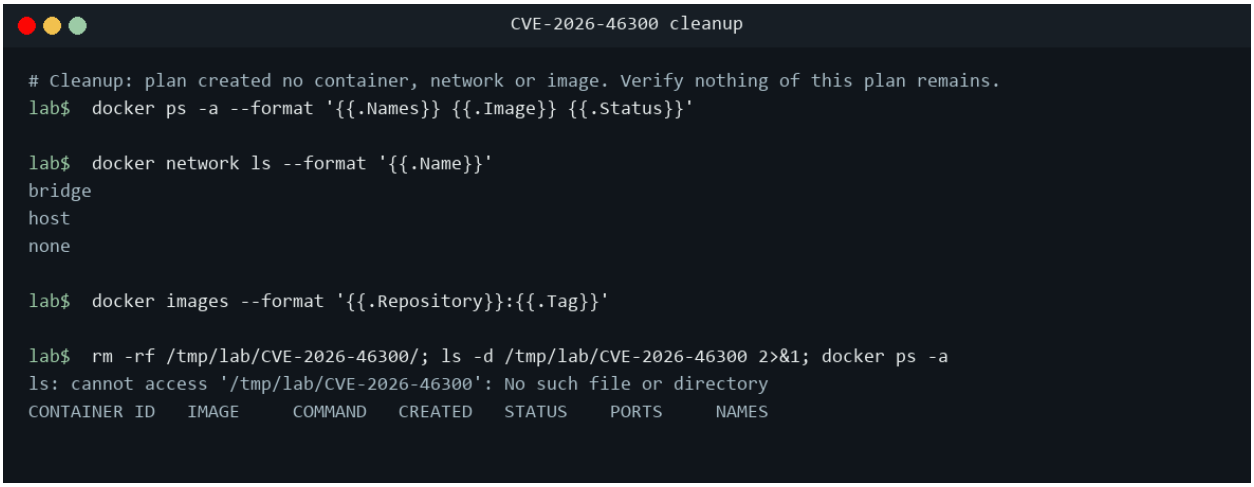
```
# Read-only verification of the plan's mitigations (nothing changed on the host)
# M1/M2 vendor/upstream fix: installed kernels and the running one
lab$  uname -r; dpkg-query -W -f='${Package} ${Version} ${Status}\n' 'linux-image-[0-9]*' | grep ' installed$'
7.1.5+kali-amd64
linux-image-6.19.14+kali-amd64 6.19.14-1+kali1 install ok installed
linux-image-7.1.5+kali-amd64 7.1.5-1kali1 install ok installed

# M4 interim module block present?
lab$  ls -l /etc/modprobe.d/fragnesia.conf 2>&1; grep -rnE '^(install|blacklist)[[:space:]]+(esp4|esp6|rxrpc)' /etc/
ls: cannot access '/etc/modprobe.d/fragnesia.conf': No such file or directory
no esp4/esp6/rxrpc block configured

lab$  modinfo -n esp4 esp6 rxrpc 2>&1
/lib/modules/7.1.5+kali-amd64/kernel/net/ipv4/esp4.ko.xz
/lib/modules/7.1.5+kali-amd64/kernel/net/ipv6/esp6.ko.xz
/lib/modules/7.1.5+kali-amd64/kernel/net/rxrpc/rxrpc.ko.xz

# M5 userns restriction
lab$  sysctl kernel.apparmor_restrict_unprivileged_userns user.max_user_namespaces 2>&1
sysctl: cannot stat /proc/sys/kernel/apparmor_restrict_unprivileged_userns: No such file or directory
user.max_user_namespaces = 30834

# M6 PHP foothold chain: is php installed on this host?
lab$  dpkg-query -W -f='${Package} ${Version}\n' 'php*' 2>&1 | head -20
php 2:8.4+99
php-amqp
php-apcu
php-ast
php-calendar
php-cgi 2:8.4+99
php-cli
php-common 2:99
php-crypt-gpg
php-ctype
php-ds
php-excimer
php-exif
php-ffi
php-fileinfo
php-fpm
php-ftp
php-gearman
php-gmagick
php-gnupg
```



(**Demonstrated**, 04-mitigation.txt and 05-cleanup.txt) [43]

Find and fix it. `uname -r` for the running kernel, then `dpkg-query -W -f='${Package} ${Version}\n' "linux-image-$(uname -r)"` (or `rpm -q kernel`), then compare with your own distribution's fixed build: for Ubuntu, the table above [9]; for Red Hat, RHSB-2026-003 [8]; for other distributions, their trackers. List every installed image, not only the running one (**Demonstrated** method) [43]. Until the fixed kernel is booted, check whether `esp4`, `esp6` and `rxrpc` are loaded or blockable, and whether unprivileged user namespaces are allowed, since the plan's interim checks looked for both (04-mitigation.txt); the vendor mitigation itself is in Red Hat's bulletin [8]. Hosts that also serve PHP are the ones KEV's chain names [4].

2.6 WordPress core, CVE-2026-119977

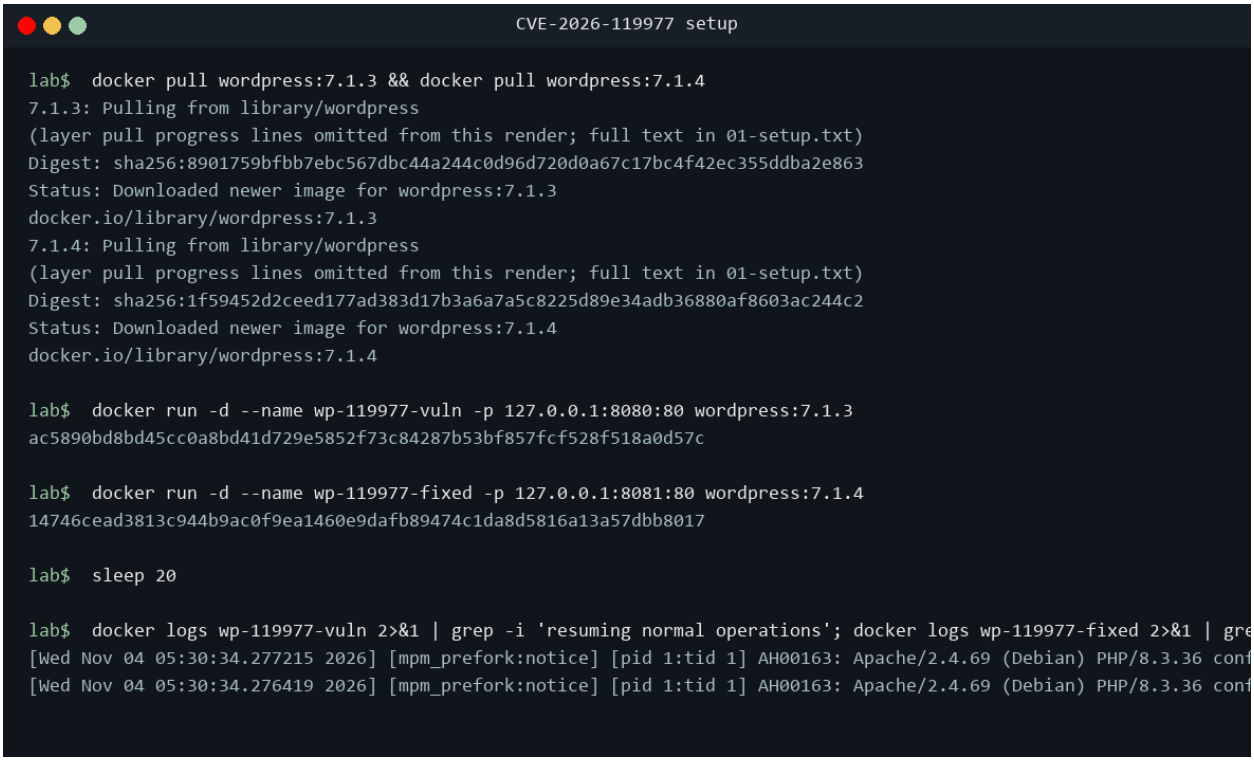
The flaw. During a full-package core update, WordPress extracted the new files into a predictable directory inside the web-served upgrade folder. An unauthenticated attacker who won the race against extraction could reach remote code execution (**Confirmed**: WordPress) [1].

The record. The vendor advisory, GHSA-4w3m-c66w-69mg, lists 25 branch ranges from 4.7.0 to 4.7.38 through 7.1.0 to 7.1.3, fixed in 7.1.4 with backports to every branch back to 4.7 (for example 7.0.8, 6.9.11 and 4.7.39), credits Umut Bayram, rates it low and carries no CVE id (**Confirmed**) [1]. WordPress 7.1.4 shipped as a security release on October 28, 2026, for a weakness in the upgrade process that can lead to code execution during the upgrade window [2]. NVD published CVE-2026-119977 on 2026-10-31, status Undergoing Analysis, with CVSS 3.1 8.1 High (`AV:N/AC:H/PR:N/UI:N`, Wordfence as secondary), CWE-367 and no CPE configuration yet; the CVE-bearing GitHub record GHSA-p6jj-qrp3-7m4g rates it high [46]. CISA listed it on 2026-11-03, due 2026-11-06, ransomware use Unknown, forensic triage Yes; SSVC exploitation active, automatable no, technical impact total [4]. No public exploit was found: the only repository a GitHub search returned was a self-described detection-only scanner created on 2026-11-04 (not read or run), and an empty CVE-named repository seen on 2026-11-03 now returns 404 [5].

Low from the vendor, high from NVD and GitHub, exploited according to CISA, and no NVD CPE yet. A scanner that keys on NVD CPE data will not flag WordPress for this CVE until NVD finishes its analysis (**Inferred** from [3]).

Why a check. No proof of concept was found, and the race exists only while a live core update runs. Showing it would mean requesting a staged installer, so the lab read `$wp_version` on 7.1.3 and 7.1.4 containers and

tested denying web access to `wp-content/upgrade/` (**Confirmed**: plan.json) [43]. The setup pulled `wordpress:7.1.3` and `wordpress:7.1.4` by tag, recorded their digests, and ran them on `127.0.0.1:8080` and `127.0.0.1:8081` (**Demonstrated**, 01-setup.txt).



What the lab showed.

```
lab$ docker exec wp-119977-vuln grep -E '^\$wp_version' /var/www/html/wp-includes/version.php
$wp_version = '7.1.3';
lab$ docker exec wp-119977-fixed grep -E '^\$wp_version' /var/www/html/wp-includes/version.php
$wp_version = '7.1.4';
```

(**Demonstrated**, 02-test.txt) Before any rule, `/wp-content/upgrade/` returned HTTP 302 on both containers, and the directory did not yet exist. After the lab created it on the vulnerable container with an `.htaccess` holding `Require all denied`, the same request returned 403, and Apache logged `AH01630: client denied by server configuration`. The fixed container, with no rule, stayed at 302 (**Demonstrated**, 02-test.txt and 04-mitigation.txt) [43]. The race itself was not exercised.

CVE-2026-119977 exposure check

```
lab$  docker exec wp-119977-vuln grep -E '^\$wp_version' /var/www/html/wp-includes/version.php
$wp_version = '7.1.3';

lab$  docker exec wp-119977-fixed grep -E '^\$wp_version' /var/www/html/wp-includes/version.php
$wp_version = '7.1.4';

lab$  docker exec wp-119977-vuln sh -c 'command -v wp || echo wp-cli not installed in image'
wp-cli not installed in image

lab$  curl -s http://127.0.0.1:8080/ | grep -io '<meta name="generator" content="WordPress [0-9.]*"' || echo 'no ger
no generator meta tag (no database configured, page does not render)

lab$  curl -s -o /dev/null -w '%{http_code}\n' http://127.0.0.1:8080/wp-content/upgrade/
403

lab$  curl -s -o /dev/null -w '%{http_code}\n' http://127.0.0.1:8081/wp-content/upgrade/
302

lab$  docker exec wp-119977-vuln ls -la /var/www/html/wp-content/upgrade/
total 12
drwxr-xr-x 2 www-data www-data 4096 Nov  4 05:31 .
drwxr-xr-x 6 www-data www-data 4096 Nov  4 05:31 ..
-rw-r--r-- 1 www-data www-data   19 Nov  4 05:31 .htaccess

lab$  docker exec wp-119977-fixed sh -c 'ls -la /var/www/html/wp-content/upgrade/ 2>&1 || true'
ls: cannot access '/var/www/html/wp-content/upgrade/': No such file or directory
```

CVE-2026-119977 mitigation check

```
lab$  docker exec wp-119977-fixed grep -E '^\$wp_version' /var/www/html/wp-includes/version.php
$wp_version = '7.1.4';

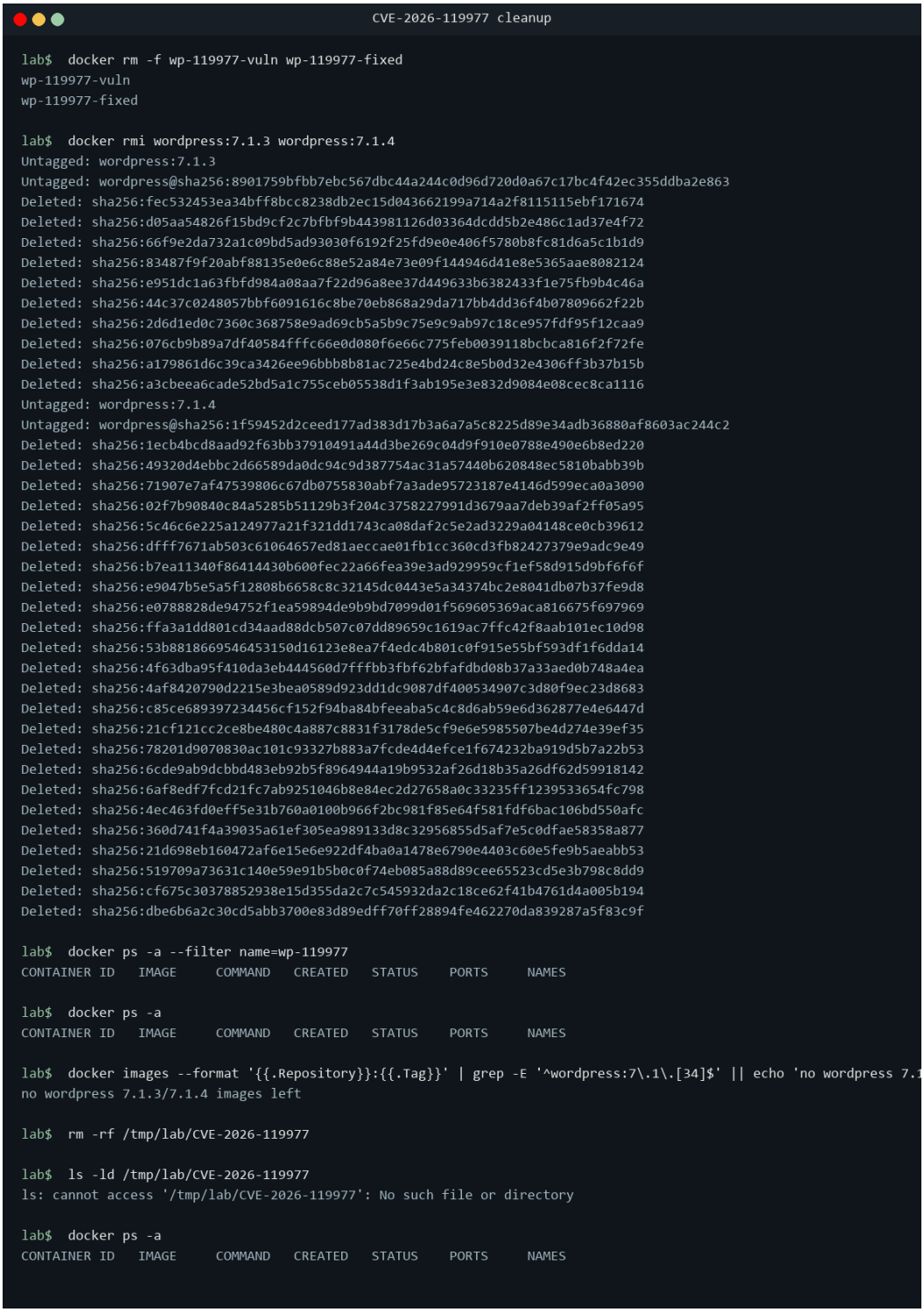
lab$  docker exec wp-119977-vuln cat /var/www/html/wp-content/upgrade/.htaccess
Require all denied

lab$  curl -s -o /dev/null -w 'vuln (deny applied) /wp-content/upgrade/ -> HTTP %{http_code}\n' http://127.0.0.1:808
vuln (deny applied) /wp-content/upgrade/ -> HTTP 403

lab$  curl -s -o /dev/null -w 'fixed (no deny) /wp-content/upgrade/ -> HTTP %{http_code}\n' http://127.0.0.1:8081/wp
fixed (no deny) /wp-content/upgrade/ -> HTTP 302

lab$  docker logs wp-119977-fixed 2>&1 | grep '/wp-content/upgrade/'
172.17.0.1 - - [04/Nov/2026:05:31:00 +0000] "GET /wp-content/upgrade/ HTTP/1.1" 302 235 "-" "curl/8.22.0"
172.17.0.1 - - [04/Nov/2026:05:31:12 +0000] "GET /wp-content/upgrade/ HTTP/1.1" 302 235 "-" "curl/8.22.0"
172.17.0.1 - - [04/Nov/2026:05:31:13 +0000] "GET /wp-content/upgrade/ HTTP/1.1" 302 235 "-" "curl/8.22.0"
```

Two version sources were missing in the images: `wp-cli` was not installed, and with no database configured the page carried no generator meta tag (**Demonstrated**, 03-exposure.txt). On a live site the generator tag may be there, but it can also be removed by a plugin, so `version.php` is the record to read (**Inferred**).



(**Demonstrated**, 05-cleanup.txt) [43]

Find and fix it. Read `$wp_version` from `wp-includes/version.php` (or run `wp core version` where wp-cli is installed) and compare it with the fixed release for that branch: 7.1.4, 7.0.8, 6.9.11 and so on back to 4.7.39 [1]. Because the window opens only during a core update, the practical bridge is to keep the upgrade folder

from being served over the web while updates run. The lab showed an Apache 2.4 `Require all denied` rule doing that on 7.1.3 (**Demonstrated**); it is not a vendor-documented workaround in the sources opened, and an Nginx equivalent was not tested (**Inferred**). Then update.

Part 3: the plans that did not finish

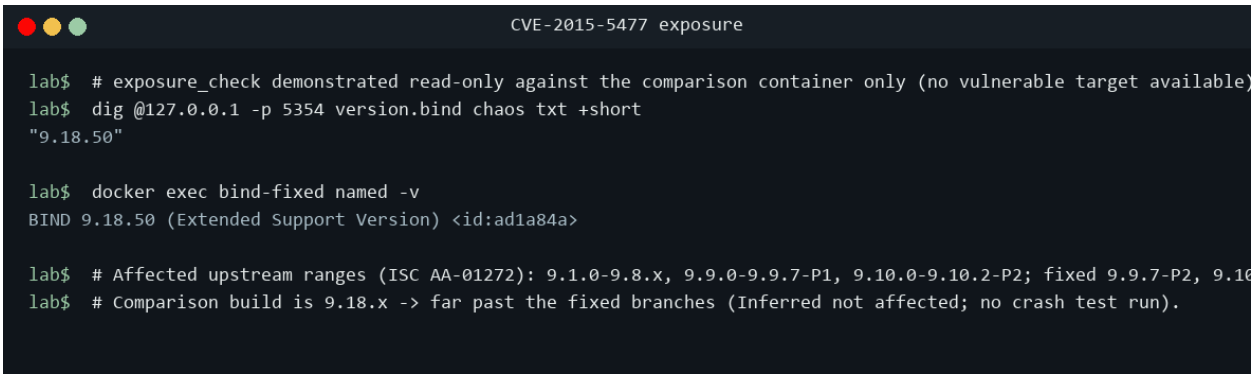
Six plans in this study's set ended blocked, inconclusive or skipped. Each one says something about the environment, not only about the CVE.

Blocked: ISC BIND, CVE-2015-5477

NVD says a remote attacker can crash `named` in BIND 9.x before 9.9.7-P2 and 9.10.x before 9.10.2-P3 with TKEY queries, through a REQUIRE assertion failure that makes the daemon exit, and lists CWE-19 and CWE-617 (**Confirmed**) [25]. ISC's AA-01272 lists 9.1.0 to 9.8.x, 9.9.0 to 9.9.7-P1 and 9.10.0 to 9.10.2-P2 as affected, 9.9.7-P2 and 9.10.2-P3 as fixed, and says there is no workaround: ACLs do not help, because the flaw is reached before ACL checks [26].

The vulnerable target never existed. Both the digest-pinned and the plain-tag pulls of `hmlio/vaas-cve-2015-5477` failed on Docker 28.5.2 with the schema 1 removal error, and the plan forbade rebuilding the image from its Dockerfile (**Demonstrated**, 01-setup.txt) [43]. Docker deprecated v2 schema 1 and Image v1 pulls in v19.03, disabled them by default in v26.0 and removed support in v28.2, and its error text matches the lab's (**Confirmed**) [27]. Many deliberately vulnerable images of this age were published in the old format; a lab on a current Docker will hit the same wall (**Inferred**).

Only the comparison container ran. It answered `version.bind` with `"9.18.50"` and `named -v` printed `BIND 9.18.50 (Extended Support Version)` (**Demonstrated**, 02-test.txt and 03-exposure.txt). No crash proof of concept was downloaded or run. That 9.18 is not affected is a reading of ISC's ranges, not a test (**Inferred**) [26].



Find and fix it. `named -v` on the host, or `dig version.bind chaos txt` from the network where the server has not hidden it, and compare with the fixed lines; patching is the only fix [26].

Blocked: Strapi, CVE-2023-22894

GitHub advisory GHSA-jjqf-j4w7-92w8 says Strapi from 3.2.1 up to but not including 4.8.0 is affected, patched in 4.8.0 [30]. NVD's text still reads "Strapi through 4.5.5" (CWE-312, CVSS 3.1 4.9), which is narrower (**Confirmed**) [31]. A check built from the NVD text alone would miss 4.6.x and 4.7.x (**Inferred**). The lab plan used the GitHub range.

Both containers logged their Strapi version, `Using strapi 4.7.1` and `Using strapi 4.8.1`, then failed with `getaddrinfo EAI_AGAIN registry.yarnpkg.com`, and both exited with code 1. No probe was sent (**Demonstrated**, 01-setup.txt) [43]. The `naskio/strapi` image runs `strapi new` in `/srv/app` on first boot when the folder is empty; its page does not mention dependency downloads, and the capture shows the package registry fetch failing [32]. The cause was container DNS: the containers used the host's private resolver, which the lab's isolation rules dropped at run time (**Demonstrated**) [43]. An image that installs itself on first boot needs network access a quarantined lab does not give it (**Inferred**).

Blocked: Citrix NetScaler, CVE-2026-88779

NVD lists NetScaler ADC before 14.1-73.41, 13.1-64.28, 14.1-73.41 FIPS and 13.1-37.282, and NetScaler Gateway before 14.1-73.41 and 13.1-64.28, as affected, CWE-119 (**Confirmed**) [34]. The CPX 14.1-73.30 image was pulled (created 2026-07-27) but never started, because the image requires `-e EULA=yes` and that acceptance is the owner's to give (**Demonstrated**, 02-test.txt) [43]. NetScaler's documentation calls EULA "a mandatory NetScaler CPX specific environment variable" and says setting it records that the user has read and accepted the End User License Agreement, with a licence-server setup as the stated exception (**Confirmed**) [33].

A verdict from the image tag alone printed:

```
14.1-73.30: BELOW fixed build 14.1-73.41 (vulnerable by version; exploitable only if SAML SP/IdP
+ Gateway/AAA configured)
```

(**Demonstrated**, 02-test.txt) The SAML precondition was not checked. On a real appliance, `show ns version` and the SAML and authentication virtual-server configuration are the two halves of the check; the lab had neither (03-exposure.txt).

Inconclusive: Apache Struts, CVE-2016-3081

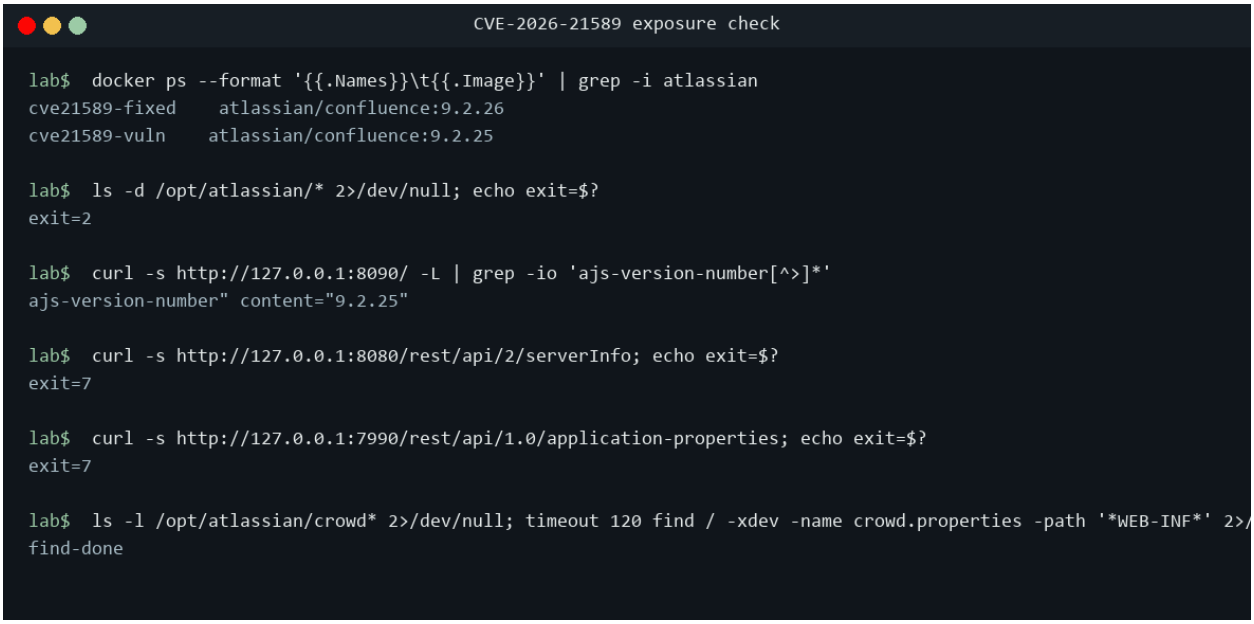
Apache's S2-032 says Struts 2.3.20 to 2.3.28 are affected except 2.3.20.3 and 2.3.24.3, only when Dynamic Method Invocation (DMI) is enabled; the fixes are 2.3.20.3, 2.3.24.3 and 2.3.28.1, and the workaround is to disable DMI (**Confirmed**) [28]. NVD gives the affected ranges as 2.3.19 to 2.3.20.2, 2.3.21 to 2.3.24.1 and 2.3.25 to 2.3.28, remote code execution through the `method:` prefix when DMI is enabled, CWE-77 [29].

The containers held `struts2-core-2.3.28.jar` and `struts2-core-2.3.30.jar` under the Maven `.m2` path, and both apps answered HTTP 200. The `struts.xml` search, filtered on `WEB-INF`, printed nothing on either. The jar's own `default.properties` reads `struts.enable.DynamicMethodInvocation = false` (**Demonstrated**, 02-test.txt) [43]. So the version half of the condition was there and the DMI half was not observed. The plan revision: search `target/classes` or the source tree without the `WEB-INF` filter, and read `struts.properties` and any DMI constant override. A default of `false` in the jar says nothing about what the application sets (**Inferred**).

Inconclusive: Atlassian Confluence, CVE-2026-21589

Atlassian's advisory says all Confluence Data Center versions are affected, the fixes are 9.2.26 and 10.2.19, no authentication is needed, and the workarounds are taking the instance off the internet, a WAF or proxy regex, or a Tomcat RewriteValve (**Confirmed**) [35]. NVD adds that exploitation requires knowing the target file's exact name and path, and that the flaw does not allow directory listing (CWE-552) [36].

`ajs-version-number` showed 9.2.25 on one container and 9.2.26 on the other (**Demonstrated**, 02-test.txt; 03-exposure.txt and 04-mitigation.txt read each one) [43].



The test step ran a public read-only detection script and three file-read requests against both containers; the script printed "probably NOT VULNERABLE" for both, and the access log recorded 404 for all three requests on both (**Demonstrated**, 02-test.txt and 04-mitigation.txt) [43]. That result is not a verdict: both instances were still in the setup wizard, and finishing it needs a licence key, an owner decision. The plan revision: complete setup with an owner-supplied licence or a pre-provisioned data volume, then rerun. The plan's log glob was also wrong: the real file is `confluence_access.YYYY-MM-DD.log` (**Demonstrated**, 04-mitigation.txt).

Skipped: Siemens Software Center, CVE-2026-110936

Its checks are Windows PowerShell file-version and registry reads, plus `sudo` on a separate Wazuh server, so nothing ran (**Demonstrated**, 03-exposure.txt) [43]. The vendor advisory was not opened for this study.

Part 4: when the version string lies

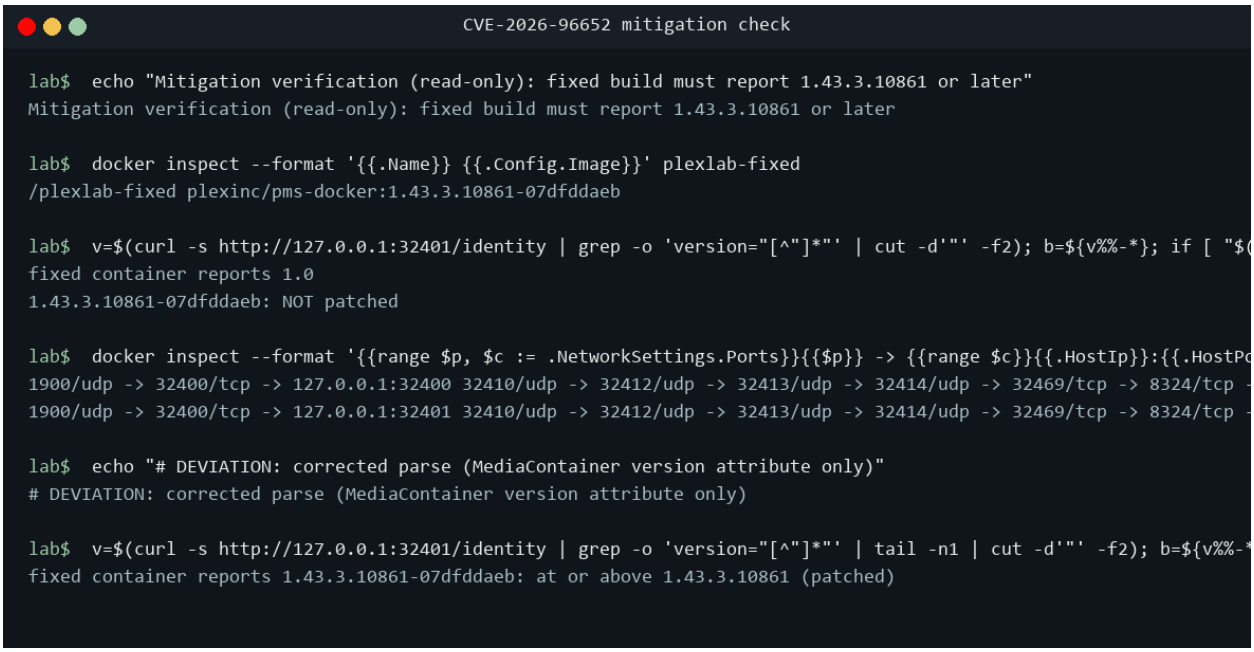
Seven checks in the same run have a home study or a short story of their own. Each is summarized here because it adds a lesson about version evidence.

- **ProFTPD, CVE-2015-3306.** The vulnerable and patched builds both report `Version: 1.3.6rc1 (git)` and both list `mod_copy.c`, so neither the banner nor `proftpd -V` can separate them. Only behavior could: the patched build answered unauthenticated `SITE CPFR` with 530 and left no marker file (**Demonstrated**, 03-exposure.txt and 04-mitigation.txt) [43]. Home study: ProFTPD mod_copy, CVE-2015-3306.
- **Node.js, CVE-2026-111497.** Node v24.22.0 and v24.22.1 both report Brotli `1.2.0`, so the bundled library's version does not separate them; the Node version does. The lab host itself showed node v24.21.0, `libbrotli1` 1.2.0-4+b1, PyPI Brotli 1.2.0 and .NET 6.0.8 (**Demonstrated**, 03-exposure.txt) [43]. Home study: Node.js, CVE-2026-111497.

- **GitLab, CVE-2026-97229.** `version-manifest.txt` read `gitlab-ce 19.4.1` and `19.4.2`, but both bundle `ruby 3.3.12 (2026-07-16 revision 0581089df9)`. The embedded Ruby string cannot tell fixed from vulnerable; the GitLab version can (**Demonstrated**, 04-mitigation.txt) [43]. Home study: the separate GitLab CVE-2026-97229 lab study.

!Terminal output on the lab host: a loop reads the first line of /opt/gitlab/version-manifest.txt in gl-vuln and gl-fixed and prints gl-vuln 19.4.1 < 19.4.2 (upgrade required) and gl-fixed 19.4.2 >= 19.4.2 (patched release); and the embedded ruby in both containers prints the same RUBY_DESCRIPTION, ruby 3.3.12 (2026-07-16 revision 0581089df9) [x86_64-linux]

- **Plex Media Server, CVE-2026-96652.** The unauthenticated `/identity` endpoint returned 1.43.2.10687 and 1.43.3.10861. But the grep for `version=` also matched the XML prolog, `version="1.0"`, so the first verdict wrongly called the fixed server "NOT patched". Adding `| tail -n1` fixed the parse. The sibling plan for CVE-2026-96654 kept the bug (**Demonstrated**, 04-mitigation.txt) [43]. Home study: Plex Media Server 2026 CVEs.



- **PHP, CVE-2026-109618 and CVE-2026-109619.** After the interim `php_admin` limits were applied (the check printed `-1 1M '0'`) and `session.upload_progress.enabled` was switched from On to Off, `X-Powered-By` still read `PHP/8.4.26`. The workaround changes configuration, not the version a scanner reads (**Demonstrated**, 04-mitigation.txt for both CVEs) [43]. Home studies: PHP 8.4.27, CVE-2026-109618 and CVE-2026-109619 for the lab run, and PHP2Shell and the October 2026 Attack Wave for the attack wave.
- **Nextcloud richdocumentscode, CVE-2026-110447.** `info.xml` showed 26.4.302 and 26.4.303, and the fix in `proxy.php` wraps the launch command in `escapeshellarg`. The `proxy.php` request count was 0 on both, so version checks leave no trace in that log (**Demonstrated**, 04-mitigation.txt) [43]. Home study: Nextcloud richdocumentscode, CVE-2026-110447.
- **Ghostcat, CVE-2020-1938.** Tomcat 9.0.30 has an active AJP connector line with no address and no secret, while stock 9.0.31 ships the AJP connector commented out and bound to `::1` (**Demonstrated**,

03-exposure.txt and 04-mitigation.txt) [43]. Here the configuration line, not the version, is the exposure (**Inferred**). The full account is in Ghostcat, CVE-2020-1938.

Part 5: the method, from the KEV feed to a per-host verdict

The lab's method can be run by any team with a shell. It is described here as the captures show it; the sequence is this study's summary (**Inferred** from the captures) [43].

1. **Build the list.** Pull the KEV JSON feed and filter `dateAdded` to the last 30 days [4].
2. **Join the records.** For each entry, take the CWE and affected ranges from NVD and the fixed builds from the vendor advisory. Where they disagree (Strapi, AhsayCBS, the WordPress severity), use the vendor's fixed build and note the disagreement [30] [31] [16] [17].
3. **Read the installed version from the most authoritative record available.** By format:

Where the software lives	Command as captured	Example in the lab
Debian and Ubuntu packages	`dpkg-query -W -f='${Package} ${Version}\n' <pkg>`, then `dpkg --compare-versions "$v" lt <fixed>`	ONLYOFFICE, kernel
RHEL-family packages	`rpm -q <pkg>`	Kernel and ONLYOFFICE checks (no RPMs on the lab host)
Containers	`docker ps --format '{{.Names}} {{.Image}}'`, `docker images --digests --no-trunc`, `docker exec <c> <version command>`	AhsayCBS
Application manifests	`wp-includes/version.php`, `/usr/local/cbs/version.txt`, `/opt/gitlab/version-manifest.txt`, `info.xml`	WordPress, AhsayCBS, GitLab, Nextcloud
Language ecosystems	`npm ls <pkg>`, `composer show`, `python3 -m pip show <pkg>`, `find / -name 'struts2-core-*.jar'`	Node.js, Struts
Network banners	`dig version.bind chaos txt`, `curl -sI` for `X-Powered-By`, `/identity`, `ajs-version-number`	BIND, PHP, Plex, Confluence
Windows	`reg query` for `CurrentBuild` and `UBR`, `Get-HotFix`, `[version]` comparison	Windows Shell
Kernel	`uname -r`, installed images via `dpkg-query`, `lsmod`, a grep of `/etc/modprobe.d/`, `sysctl` for user namespaces	Fragnesia

(**Demonstrated**: the commands as they appear in the captures) [43]

4. **Settle the verdict with a version-aware comparison.** `printf '%s\n%s\n' "$v" <fixed> | sort -V | head -n1` on Linux, `dpkg --compare-versions` for Debian packages, `[version]` in PowerShell. Never compare version strings as text (**Inferred**).
5. **Read the precondition line.** JWT for ONLYOFFICE, DMI for Struts, SAML for NetScaler, the AJP connector for Tomcat, the statistics feature for SiteGuard, the loaded modules for Fragnesia [23] [28] [43] [11].

6. **Check the workaround separately.** A deny rule, a module block or an interim limit is a configuration fact with its own evidence; record it next to the version finding, not in place of it [43] (**Inferred**).

The traps the lab hit, each one a capture [43]:

Trap	What happened	Capture
Same banner on both builds	ProFTPD, Node.js Brotli, GitLab Ruby	CVE-2015-3306, CVE-2026-111497, CVE-2026-97229
A parser matching the wrong field	Plex prolog `version="1.0"`	CVE-2026-96652 04-mitigation.txt
A regex that never matched	Minified `api.js`	CVE-2021-3199 02-test.txt
A process check matching itself	`pgrep -f -i siteguard`	CVE-2026-106931 02-test.txt
A threshold from the wrong distribution	Ubuntu line on a Kali kernel printed `FIXED`	CVE-2026-46300 02-test.txt
A wrong log path	`conf_access_log*.log` did not exist	CVE-2026-21589 04-mitigation.txt
A default that is not the setting	Struts jar default for DMI	CVE-2016-3081 02-test.txt

Part 6: the vulnerability-management view

- **The listing date is not the flaw's age.** Five CVEs from 2015 to 2023 arrived on one day with a three-day clock [4]. Old software in an inventory is not a solved problem because its CVE is old; ONLYOFFICE CVE-2021-3199 waited more than five years for its KEV entry [23] [4] (**Inferred**).
- **Severity sources disagree, and the clock wins.** WordPress's own advisory says low, NVD carries 8.1 High, GitHub's CVE record says high, and CISA says exploited with a three-day due date [1] [3] [4] [46]. For a KEV entry the due date and the SSVC "active" decide the priority; the base score decides little (**Inferred**).
- **Missing CPE data blinds CPE-keyed scanners.** NVD had no CPE configuration for CVE-2026-119977 on 4 November 2026 [3]. Until it does, a scanner that matches NVD CPEs to installed software will not raise the finding, and a team that waits for its scanner will miss the due date (**Inferred**).
- **Incomplete fixes create a false "patched" state.** Ahsay's first patches, 10.3.4.0 and 9.15.2.59, did not fully fix either CVE [15]. Tickets closed against the first hotfix need reopening; check the build number, not the change record (**Inferred**).
- **Forensic triage is part of the fix.** CISA marks forensic triage Yes for six of the seven entries whose field this study records (AhsayCBS CVE-2026-105133 is No) [4], and SiteGuard warns that patching may erase traces [11]. Collect the evidence first, then patch.
- **Edge devices are where the lab is weakest.** The appliance entries in the window had no lab target [4]. A container lab is good evidence for software you can run in a container and weak evidence for the devices that most often face the internet (**Inferred**).

The library's CVE Reference is the place to look up the older CVEs in this list; whether it has entries for each of the 2026 CVEs was not checked for this study.

Overstated and understated

- **Overstated:** a version match as proof of exposure. ONLYOFFICE needs JWT and Struts needs DMI (**Confirmed** [23] [28]); the lab plan reads a SAML precondition into NetScaler, and the Ghostcat pair differed in its AJP connector line, not its banner (**Demonstrated** [43]). A scanner finding on version alone overstates the exposed set (**Inferred**). Also overstated: a base score as the measure of urgency, when the vendor rates WordPress CVE-2026-119977 low, NVD carries 8.1 High and CISA gave it three days (**Confirmed** [1] [3] [4]); and a check's own "FIXED" line, which on the Kali kernel compared against Ubuntu's threshold and meant nothing (**Demonstrated** [43]).
- **Understated:** old flaws, since five CVEs from 2015 to 2023 arrived on one day with a three-day clock (**Confirmed** [4]); the first AhsayCBS hotfixes, which Ahsay itself says did not fully fix either CVE (**Confirmed** [15]); the fallback kernel image that sits inside an affected range while the running one does not (**Inferred** from [6] and the lab reading [43]); a mitigated host that a version-keyed scanner still flags (**Demonstrated** [43]); and the edge appliances in the window, which no container lab reached (**Confirmed** [4]).

MITRE mappings

No ATT&CK procedure example in the opened record names any of these CVEs. Each technique below is this study's binding of a flaw's documented attack path to the technique's definition, with the basis stated. Every id was checked against the library's `data/attack/technique_profiles.jsonl` on main, which has a record and a generated page for each, and T1190 and T1499.004 were also opened on the live ATT&CK site [37] [38] [40].

ATT&CK Enterprise techniques

ATT&CK id	Name	Tactic (v20)	Dated event or report	Basis	Source
T1190	Exploit Public-Facing Application	Initial Access	ONLYOFFICE `/upload` traversal with RCE (NVD 2021-01-26); AhsayCBS API bypass and Replication Receiver RCE, remote and unauthenticated (NVD 2026-10-04); Confluence unauthenticated file access (Atlassian 2026-10-05); WordPress unauthenticated upgrade race (GHSA 2026-10-29); Struts DMI RCE (NVD 2016-04-26)	Remote, unauthenticated flaws in internet-facing server software (Confirmed records); the lab showed the vulnerable version on matched ONLYOFFICE, AhsayCBS and WordPress targets, and the JWT precondition on ONLYOFFICE (Demonstrated)	[23] [14] [16] [35] [1] [29] [43] [37]
T1499.004	Application or System Exploitation	Impact	BIND TKEY assertion failure and daemon exit (ISC 2015-07-28, NVD 2015-07-29); NetScaler memory-safety flaw, CWE-119 (NVD 2026-10-04)	The BIND crash is the record's stated impact (Confirmed); the NetScaler binding is Inferred from the lab plan	[26] [25] [34] [38]
T1068	Exploitation for Privilege Escalation	Privilege Escalation	Fragnesia: a low-privileged local attacker overwrites privileged binaries for root (Red Hat; oss-security 2026-05-13); SiteGuard: local vector `AV:L/PR:L` with commands as root or SYSTEM (JVN 2026-10-14)	Fragnesia Confirmed; SiteGuard Inferred from the vector and impact	[8] [7] [10] [12] [40]
T1505.003	Web Shell	Persistence	KEV: Fragnesia chained with CVE-2026-109618 and CVE-2026-109619 to implant PHP web shells (2026-10-28); Huntress: JSP web shells on AhsayCBS (2026-10-07, Reported); Horizon3: JSP write (Reported)	KEV text Confirmed; the AhsayCBS accounts Reported	[4] [18] [17] [40]
T1059.004	Unix Shell	Execution	SiteGuard OS command injection on the Apache and Nginx editions (JVN 2026-10-14)	Inferred: the record says OS commands, not which shell	[10] [11] [40]
T1105	Ingress Tool Transfer	Command and Control	Huntress: XMRig miners placed on exploited AhsayCBS servers (2026-10-07 to 2026-10-08)	Reported	[18] [40]
T1496.001	Compute Hijacking	Impact	Huntress: XMRig miners on AhsayCBS (2026-10-07 to 2026-10-08)	Reported	[18] [40]
T1553.005	Mark-of-the-Web Bypass	Defense Impairment	Microsoft: a successful exploit evades Mark of the Web (2026-10-13)	Confirmed record	[20] [40]

ATT&CK id	Name	Tactic (v20)	Dated event or report	Basis	Source
T1566.001	Spearphishing Attachment	Initial Access	Microsoft: the attacker sends a crafted file by email or instant message (2026-10-13)	Confirmed record; the email path	[20] [40]
T1204.002	Malicious File	Execution	Microsoft: the user must click a link or open the file (2026-10-13)	Confirmed record	[20] [40]
T1005	Data from Local System	Collection	Confluence arbitrary file access with a known file name and path (Atlassian and NVD, 2026-10-05)	Inferred binding of CWE-552 file access	[35] [36] [40]
T1595.002	Vulnerability Scanning	Reconnaissance	The banner reads the lab used (`version.bind`, `X-Powered-By`, `/identity`, `ajs-version-number`) are the reads a scanner makes (lab, 2026-11-03 to 2026-11-04)	Inferred: what defenders read, an attacker can read	[43] [40]

T1083 File and Directory Discovery is left out: NVD says the Confluence flaw does not allow directory listing [36]. The Siemens Software Center flaw is not bound, because its advisory was not opened for this study. ATT&CK for ICS is not bound: NVD lists Siemens SIMATIC firmware for Fragnesia, but the record describes no ICS-specific behavior (**Inferred**) [6]. ATT&CK for Mobile does not apply.

ATT&CK lists these mitigations on T1190, and the library has a page for each: M1051 Update Software, M1016 Vulnerability Scanning, M1037 Filter Network Traffic and M1030 Network Segmentation [37] [40]. The method in Part 5 is M1016 and M1051 done carefully.

Group, software and campaign objects

None is bound. The KEV entries record ransomware use as Unknown and name no actor [4], the vendor records opened name none, and the only named tool in the record, the XMRig miner in Huntress's AhsayCBS report, has no ATT&CK software object in the library's data, so none is bound [18].

CWE and CAPEC

CVE	CWE (as the record lists it)	CAPEC (library mapping)	Source
CVE-2026-119977 WordPress	CWE-367 (NVD)	CAPEC-29 Leveraging Time-of-Check and Time-of-Use (TOCTOU) Race Conditions	[3] [41]
CVE-2026-46300 Fragnesia	CWE-787 (NVD), CWE-456 (CISA ADP and KEV), CWE-123	none bound	[6] [4]
CVE-2026-106931 SiteGuard	CWE-78 (JVN)	CAPEC-88 OS Command Injection	[10] [41]
CVE-2026-105133 AhsayCBS	CWE-287	none bound	[14] [41]
CVE-2026-105134 AhsayCBS	CWE-77, CWE-78	CAPEC-88 OS Command Injection (through CWE-78)	[16] [41]
CVE-2026-96951 Windows Shell	CWE-77	none bound	[4]
CVE-2021-3199 ONLYOFFICE	CWE-22	CAPEC-126 Path Traversal	[23] [41]
CVE-2015-5477 BIND	CWE-19 and CWE-617 (NVD); KEV lists CWE-19	none bound	[25] [4] [41]
CVE-2016-3081 Struts	CWE-77	none bound	[29]
CVE-2023-22894 Strapi	CWE-312	CAPEC-37 Retrieve Embedded Sensitive Data	[31] [41]
CVE-2026-88779 NetScaler	CWE-119	none bound	[34]
CVE-2026-21589 Confluence	CWE-552	CAPEC-639 Probe System Files	[36] [41]
CVE-2015-3306 ProFTPD	CWE-284 (KEV)	none bound	[4]

Each CAPEC row follows the library's `related_cwe` link in `data/weaknesses/capec.jsonl`; that each attack pattern fits the flaw is this study's reading (**Inferred**) [41]. The library's CWE data has CWE-617 and not CWE-19, so the BIND row's library reference is CWE-617 [41]. The library has no per-id CWE pages; see the CWE Reference.

D3FEND countermeasures

Countermeasure	Id	Why it applies here	Library mapping
Software Update	D3-SU	Replacing old software, the class every fixed build in this study belongs to [39]	Page exists; not mapped to T1190 in the library [42]
Asset Vulnerability Enumeration	D3-AVE	The version-against-fixed-build comparison in Part 5	Page exists; not mapped to T1190 [42]
Software Inventory	D3-SWI	Finding every copy: packages, containers, manifests	Page exists [42]
Inbound Traffic Filtering	D3-ITF	Keeping AhsayCBS `/rps/`, Confluence and the WordPress upgrade path off the internet while patching	Mapped to T1190 [42]
Network Traffic Filtering	D3-NTF	As above, between internal tiers	Mapped to T1190 [42]
Application Protocol Command Analysis	D3-APCA	Inspecting requests to the vulnerable endpoints	Mapped to T1190 [42]
File Integrity Monitoring	D3-FIM	Finding planted JSP and PHP web shells; the cached-versus-direct hash of `/usr/bin/su` is the same idea for the page cache	Mapped to T1505.003 [42]
Executable Denylisting	D3-EDL	Blocking known miner binaries on exploited servers	Mapped to T1505.003 [42]

The mappings are the library's; the "why it applies" column is this study's reading (**Inferred**). The library has no D3FEND mapping for T1499.004 [42]. See the D3FEND Reference.

Navigator layer

navigator/library-layers/exposure-checks-2026-11.json carries the twelve technique entries above (ATT&CK v20, domain `enterprise-attack`); see Navigator layers for how to open it. Score 100 marks a binding the record supports and the lab backed with a vulnerable version on a matched target (T1190); 75 marks a binding from a Confirmed record without a lab demonstration; 50 marks a binding from a Reported source or this study's own inference. Each entry's comment names the source numbers.

MITRE ATT&CK, CWE, CAPEC and D3FEND content is used under the terms recorded in THIRD_PARTY_NOTICES.md.

Cross-cutting themes

- **Version plus precondition is the unit of exposure.** Four of the flaws in this study turn on one configuration line, and the lab's clearest verdicts came from reading both halves (**Demonstrated**, with the reading **Inferred**) [43]. For the related problem of a bundled library hidden behind a product's own version, see Log4Shell.
- **The record is several records.** Vendor, NVD, GitHub and CISA texts disagreed on Strapi's range, AhsayCBS's mechanism and WordPress's severity [30] [31] [16] [17] [1] [3] [4]. A check that cites one record should say which, and why (**Inferred**).

- **Safe labs produce honest gaps.** Not rebuilding an image the plan forbade, not opening the lab's egress for a first-boot installer, and not accepting a licence on the owner's behalf left three plans blocked (BIND, Strapi, NetScaler) and Confluence inconclusive, each with its reason written down; Struts stayed inconclusive because the DMI setting it needed was not found [43]. A gap with a reason is evidence too (**Inferred**).
- **The checker is code, and code has bugs.** The Plex prolog match, the minified `api.js` regex and the self-matching `pgrep` were all bugs in the checks, not in the targets [43]. Test your checks on a known-fixed and a known-vulnerable target before you trust them across a fleet, which is what the matched pairs did here (**Inferred**).
- **The edge is still the blind spot.** The appliance entries the lab could not reach are the ones most likely to face the internet [4]. For those, the vendor's own version command and the vendor's fixed-build list are the whole method, and they need to be run on the device (**Inferred**). The Linux TCP use-after-free study makes the same point for kernels.

Sources

Sources 1 to 16, 19 to 42, 45 and 46 were opened on 4 November 2026. Sources 17 and 18 were opened on 3 November 2026 and not re-opened. Sources 43 and 44 are the lab captures and the run report, recorded on 3 and 4 November 2026.

#	Source	URL	Opened	Grade
1	WordPress security advisory GHSA-4w3m-c66w-69mg (published 2026-10-29)	https://github.com/WordPress/wordpress-develop/security/advisories/GHSA-4w3m-c66w-69mg	2026-11-04	Confirmed
2	WordPress 7.1.4 security release (October 28, 2026)	https://wordpress.org/news/2026/10/wordpress-7-1-4/	2026-11-04	Confirmed
3	NVD record CVE-2026-119977 (API 2.0)	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2026-119977	2026-11-04	Confirmed
4	CISA Known Exploited Vulnerabilities feed (catalog version 2026.11.03)	https://www.cisa.gov/sites/default/files/feeds/known_exploited_vulnerabilities.json	2026-11-04	Confirmed
5	GitHub repository search for CVE-2026-119977	https://api.github.com/search/repositories?q=CVE-2026-119977	2026-11-04	Confirmed
6	NVD record CVE-2026-46300 (API 2.0)	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2026-46300	2026-11-04	Confirmed
7	oss-security, Fragnesia disclosure thread (2026-05-13)	https://www.openwall.com/lists/oss-security/2026/05/13/5	2026-11-04	Confirmed
8	Red Hat, CVE-2026-46300	https://access.redhat.com/security/cve/CVE-2026-46300	2026-11-04	Confirmed
9	Ubuntu security, CVE-2026-46300 (JSON)	https://ubuntu.com/security/cves/CVE-2026-46300.json	2026-11-04	Confirmed
10	JVN#88474777, SiteGuard Server Edition	https://jvn.jp/en/jp/JVN88474777/	2026-11-04	Confirmed
11	EG Secure Solutions, SiteGuard Server Edition alert (2026-10-14, Japanese)	https://www.eg-secure.co.jp/siteguard/news/alert-202610-1	2026-11-04	Confirmed

#	Source	URL	Opened	Grade
12	NVD record CVE-2026-106931 (API 2.0)	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2026-106931	2026-11-04	Confirmed
13	GitHub repository search for CVE-2026-106931	https://api.github.com/search/repositories?q=CVE-2026-106931	2026-11-04	Confirmed
14	NVD record CVE-2026-105133 (API 2.0; CNA VulDB)	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2026-105133	2026-11-04	Confirmed
15	Ahsay announcement, patch update for CVE-2026-105133 and CVE-2026-105134 (October 12, 2026)	https://www.ahsay.com/en/company/announcements/patch-cve-105133-105134-update	2026-11-04	Confirmed
16	NVD record CVE-2026-105134 (API 2.0; CNA VulDB)	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2026-105134	2026-11-04	Confirmed
17	Horizon3.ai, AhsayCBS CVE-2026-105133 and CVE-2026-105134 technical analysis	https://horizon3.ai/attack-research/attack-blogs/ahsaycbs-cve-2026-105133-cve-2026-105134-technical-analysis/	2026-11-03	Reported
18	Huntress, AhsayCBS exploitation report	https://www.huntress.com/blog/ahsaycbs-flaws-exploit	2026-11-03	Reported
19	GitHub API, horizon3ai/AhsayCBS-Auth-Bypass-RCE repository metadata	https://api.github.com/repos/horizon3ai/AhsayCBS-Auth-Bypass-RCE	2026-11-04	Confirmed (metadata only)
20	Microsoft Security Update Guide API, CVE-2026-96951	https://api.msrc.microsoft.com/sug/v2.0/en-US/vulnerability?$filter=cveNumber%20eq%20%27CVE-2026-96951%27	2026-11-04	Confirmed
21	NVD record CVE-2026-96951 (API 2.0)	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2026-96951	2026-11-04	Confirmed
22	GitHub repository search for CVE-2026-96951	https://api.github.com/search/repositories?q=CVE-2026-96951	2026-11-04	Confirmed
23	NVD record CVE-2021-3199 (API 2.0)	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2021-3199	2026-11-04	Confirmed
24	ONLYOFFICE DocumentServer CHANGELOG at commit 903fe5a	https://github.com/ONLYOFFICE/DocumentServer/blob/903fe5ab7a275bd69c3c3346af2d21cf87ebeabf/CHANGELOG.md	2026-11-04	Confirmed
25	NVD record CVE-2015-5477 (API 2.0)	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2015-5477	2026-11-04	Confirmed
26	ISC Knowledge Base AA-01272	https://kb.isc.org/docs/aa-01272	2026-11-04	Confirmed
27	Docker Engine, deprecated features	https://docs.docker.com/engine/deprecated/	2026-11-04	Confirmed
28	Apache Struts security bulletin S2-032	https://cwiki.apache.org/confluence/display/WW/S2-032	2026-11-04	Confirmed
29	NVD record CVE-2016-3081 (API 2.0)	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2016-3081	2026-11-04	Confirmed
30	GitHub Advisory GHSA-jjqf-j4w7-92w8 (Strapi)	https://github.com/advisories/GHSA-jjqf-j4w7-92w8	2026-11-04	Confirmed

#	Source	URL	Opened	Grade
31	NVD record CVE-2023-22894 (API 2.0)	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2023-22894	2026-11-04	Confirmed
32	Docker Hub, naskio/strapi	https://hub.docker.com/r/naskio/strapi	2026-11-04	Confirmed
33	NetScaler CPX documentation, deploy using the Docker image file	https://docs.netscaler.com/en-us/cpx/current-release/deploy-using-docker-image-file	2026-11-04	Confirmed
34	NVD record CVE-2026-88779 (API 2.0)	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2026-88779	2026-11-04	Confirmed
35	Atlassian security advisory, CVE-2026-21589	https://confluence.atlassian.com/security/cve-2026-21589-arbitrary-file-access-vulnerability-impacts-multiple-products-1870495748.html	2026-11-04	Confirmed
36	NVD record CVE-2026-21589 (API 2.0)	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2026-21589	2026-11-04	Confirmed
37	MITRE ATT&CK T1190 (ATT&CK v20, technique version 2.8)	https://attack.mitre.org/techniques/T1190/	2026-11-04	Confirmed
38	MITRE ATT&CK T1499.004 Application or System Exploitation, a sub-technique of T1499 Endpoint Denial of Service (version 1.3, last modified 24 October 2025)	https://attack.mitre.org/techniques/T1499/004/	2026-11-04	Confirmed
39	MITRE D3FEND, Software Update (D3-SU)	https://d3fend.mitre.org/technique/d3f:SoftwareUpdate/	2026-11-04	Confirmed
40	Library data on main: `data/attack/technique_profiles.jsonl` and the `mitre/techniques/` and `mitre/mitigations/` pages	https://github.com/TeamStarWolf/TeamStarWolf/blob/main/data/attack/technique_profiles.jsonl	2026-11-04	Confirmed
41	Library data on main: `data/weaknesses/capec.jsonl` and `data/weaknesses/cwe.jsonl`, and the `mitre/capec/` pages	https://github.com/TeamStarWolf/TeamStarWolf/blob/main/data/weaknesses/capec.jsonl	2026-11-04	Confirmed
42	Library data on main: `data/attack/technique_to_d3fend.jsonl` and the `mitre/d3fend/` pages	https://github.com/TeamStarWolf/TeamStarWolf/blob/main/data/attack/technique_to_d3fend.jsonl	2026-11-04	Confirmed
43	Lab captures: `plan.json`, `01-setup.txt`, `02-test.txt`, `03-exposure.txt`, `04-mitigation.txt` and `05-cleanup.txt` with their renders, for each CVE named in this study	not published	2026-11-04	Demonstrated
44	Lab run report, leftovers check (3 to 4 November 2026)	not published	2026-11-04	Demonstrated (not re-checked)
45	GitHub Advisory GHSA-vm8r-r39c-96c6 (ONLYOFFICE)	https://github.com/advisories/GHSA-vm8r-r39c-96c6	2026-11-04	Confirmed
46	GitHub Advisory GHSA-p6jj-qrp3-7m4g (WordPress)	https://github.com/advisories/GHSA-p6jj-qrp3-7m4g	2026-11-04	Confirmed

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Related references

- CVE Reference, CWE Reference, CAPEC Reference, D3FEND Reference
- Threat Actors, Malware Families, Notable Incidents: no actor or malware family is attributed to the exploitation in this study
- Vulnerability Prioritization Reference, Vulnerability Management Reference, Triage a CVE
- Internet-Facing Services Reference, Container Security Reference, Debian, Ubuntu, Windows
- Lab-backed studies summarized in Part 4: ProFTPD mod_copy; Node.js Brotli decoder; Plex Media Server Plex2Shell; PHP 8.4.27; Collabora Built-in CODE Server in Nextcloud; Ghostcat, the Tomcat AJP file read in the lab; and the other lab study from the same run, Magento and Adobe Commerce PolyShell
- Other vulnerability studies: PHP2Shell and the October 2026 attack wave; Log4Shell, a bundled library hidden behind a product's own version; the Linux TCP use-after-free and the Multipath TCP attack surface, two more kernel readings from the same lab host
- Techniques: T1190 Exploit Public-Facing Application, T1499.004 Application or System Exploitation, T1068 Exploitation for Privilege Escalation, T1505.003 Web Shell, T1059.004 Unix Shell, T1105 Ingress Tool Transfer, T1496.001 Compute Hijacking, T1553.005 Mark-of-the-Web Bypass, T1566.001 Spearphishing Attachment, T1204.002 Malicious File, T1005 Data from Local System, T1595.002 Vulnerability Scanning
- Mitigations: M1051 Update Software, M1016 Vulnerability Scanning, M1037 Filter Network Traffic, M1030 Network Segmentation
- CAPEC: CAPEC-29 Leveraging Time-of-Check and Time-of-Use (TOCTOU) Race Conditions, CAPEC-88 OS Command Injection, CAPEC-126 Path Traversal, CAPEC-37 Retrieve Embedded Sensitive Data, CAPEC-639 Probe System Files
- D3FEND: Software Update, Asset Vulnerability Enumeration, Software Inventory, Inbound Traffic Filtering, File Integrity Monitoring, all countermeasures
- Navigator layers, including this study's layer, exposure-checks-2026-11.json
- All case studies